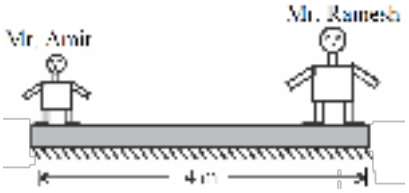
PHYSICS

1) An alpha particle is accelerated through a potential difference of 100V. To have same de-Broglie wavelength what potential difference must be applied across singly ionised ${}_4Be^8$ atom :-

(1) 200 V
(2) 25 V
(3) 100 V
(4) 50 V

2) Mr. Ramesh (60 kg) and Mr. Amit (50 kg) are sitting at the two extremes of a 4 m long boat (40 kg) standing still in water. They come to middle of boat to talk. Neglecting friction with water, how far does the boat moves on the water during the process ?



(1) $\frac{3}{15}$m
(2) $\frac{2}{15}$m
(3) $\frac{4}{15}$m
(4) None of these

3) A car starts moving along a line, first with acceleration a = 5 m/s^2 starting from rest then uniformly and finally decelerating at the same rate and comes to rest. The total time of motion is 25 s. The average speed during the time is 20 m/s. How long does the particle move uniformly ?

(1) 10 s
(2) 12 s
(3) 20 s
(4) 15 s

4) A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x, the moment of inertia is given by I = $2x^2$ – 12x + 27. The x-coordinate of its centre of mass is

(1) x = 2
(2) x = 0
(3) x = 1
(4) x = 3

5) **Statement-I:** A spinning cricket ball moving through air, does not deviate from its normal

trajectory.
Statement-II: Magnus effect is an application of Bernoulli's principle.

(1) Both statement-I and statement- II are true
(2) Both statement-I and statement- II are false
(3) Statement-I is true and statement-II is false
(4) Statement-I is false and statement-II is true

6) In a series L-C-R circuit, current in the circuit is 11 A when the applied voltage is 220 V. Voltage across the capacitor is 200 V. If value of resistor is 20 Ω, then the voltage across the unknown inductor is :-

(1) zero
(2) 200 V
(3) 20 V
(4) None of these

7) A force of 50 dyne is acting on a body at rest having mass 5g for an interval of 3 seconds, then impulse is :

(1) $0.15 \times 10^{-3} Ns$
(2) $0.98 \times 10^{-3} Ns$
(3) $1.5 \times 10^{-3} Ns$
(4) $2.5 \times 10^{-3} Ns$

8) The binding energy per nucleon of C-12 is 7.68 MeV and C-13 is 7.48 MeV. The energy (in MeV) required to remove the extra neutron from C-13 is very nearly equal to :-

(1) 0.2
(2) 3.7
(3) 3.9
(4) 5

9)

A doubly ionized lithium atom is hydrogen like with atomic number 3. Find the wavelength of radiation required to excite electron in Li^{2+} from first to third bohr orbit. (hc = 1240 eVnm)

(1) 11.4 nm
(2) 10.7 nm
(3) 12.7 nm
(4) 13.4 nm

10) Work function is :-

(1) Energy necessary to eject the electron from its orbit
(2) Energy necessary to eject the electron from metal

(3) Minimum necessary energy to eject the electron from metal

(4) Wavelength necessary for releasing an electron from a body

11) Column-I specifics a point distance r from the center/axis of a symmetrical distribution of charge. Column-II gives the variation of electric field intensity at P as a function of r. Match the columns :-

	Column-I		**Column-II**
i	P lies outside a straight wire having uniform linear charge density	a	$E \propto \frac{1}{r^2}$
ii	P lies inside a spherical charged conductor	b	$E \propto \frac{1}{r}$
iii	P lies inside a spherical body having uniform volume charge density	c	$E \propto r$
iv	P lies outside the plane uniformly charged non-conducting sheet	d	$E \propto r^0$

(1) i-b, ii-d, iii-c, iv-d

(2) i-a, ii-b, iii-c, iv-d

(3) i-b, ii-d, iii-a, iv-c

(4) i-d, ii-b, iii-a, iv-c

12) For refraction through a small angled prism, the angle of deviation (Refractive indices : $n_{surrounding} < n_{prism}$).
(a) increases with the increase in refractive index of the prism.
(b) will be doubled if refractive index of the prism is doubled.
(c) is directly proportional to the angle of the prism.
(d) will decrease with the increase in refractive index of the prism.
Which statements are correct :

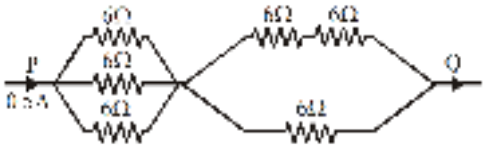
(1) a and b

(2) a, b and c

(3) a and c

(4) d only

13) Resistances of 6 ohm each are connected in the manner shown in adjoining figure. With the current 0.5 ampere as shown in figure, the potential difference $V_P - V_Q$ is :-



(1) 3.6 V

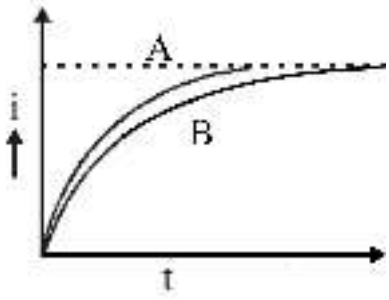
(2) 6.0 V

(3) 3.0 V

(4) 7.2 V

14) A circuit consisting of a constant emf E, a self-inductance L and a resistance R is closed at time t = 0. The relation between the current i in the circuit and the time t is as shown by the curve 'A' in

the figure. When one or more of parameters E, R and L are changed, the curve 'B' is obtained. Then it is possible that :-



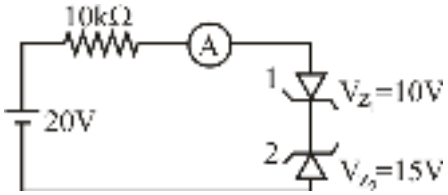
(1) E and R are kept constant and L is increased

(2) E and R are kept constant and L is decreased

(3) E and L are kept constant and R is decreased

(4) None of these

15) The reading of ammeter in the following circuit will be :-



(1) 0.5 mA

(2) 1.0 mA

(3) 1.5 mA

(4) 2.5 mA

16) Two sources of waves are called coherent if :-

(1) Both have the same amplitude of vibrations

(2) Both produce waves of the same wavelength

(3) Both produce waves of the same wavelength having constant phase difference

(4) Both produce waves having the same velocity

17) The gas mixture consists of 3 moles of oxygen and 5 moles of argon at temperature T. Considering only translational and rotational modes, the total internal energy of system is-

(1) 12 RT

(2) 20RT

(3) 15RT

(4) 4RT

18) Each plate of a parallel plate air capacitor has given charge 'q', surface area of each plate is A & plate separation is 'd' then :-

(1) Capacitance will be zero

(2) Energy stored between plates is $\frac{q^2A}{\in_0 d}$

(3) No charge on inner surface of each plate

(4) Potential difference between the plates is $\frac{2qd}{\in_0 A}$

19) If two SHM are represented by equations
$y_1 = 10\sin[8\pi t + \pi/3]$
$y_2 = 5[\sin 8\pi t + \cos 8\pi t]$
then the phase difference between them, is

(1) $\frac{\pi}{12}$ rad

(2) $\frac{\pi}{6}$ rad

(3) $\frac{\pi}{2}$ rad

(4) $\frac{\pi}{4}$ rad

20) A vector has magnitude same as that of $\vec{A} = 3\hat{i} + 4\hat{j}$ and is parallel to $\vec{B} = 4\hat{i} + 3\hat{j}$. The x and y components of this vector in first quadrant are x and 3 respectively where x is

(1) 5

(2) $\frac{4}{5}$

(3) $\frac{3}{5}$

(4) 4

21) The coefficient of friction between a body and the surface of an inclined plane at 45° is 0.5. If g = 9.8 m/s^2, the downward acceleration of the body in m/s^2 is-

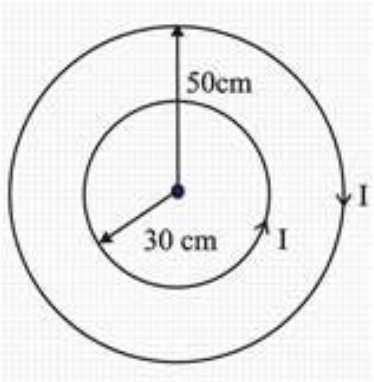
(1) $\frac{4.9}{\sqrt{2}}$

(2) $4.9\sqrt{2}$

(3) $19.6\sqrt{2}$

(4) 4.9

22) Two concentric circular loops of radii r_1=30 cm and r_2=50 cm are placed in X-Y plane as shown in the figure. A current I = 7A is flowing through them in the direction as shown in figure. The net



magnetic moment of this system of two circular loops is approximately :

(1) $\frac{7}{2}\hat{k}\ Am^2$

(2) $-\frac{7}{2}\hat{k}\ Am^2$

(3) $7\hat{k}\ Am^2$

(4) $-7\ \hat{k}\ Am^2$

23) A bi–convex lens has radius of curvature of both the surfaces same as 1/6 cm. If this lens is required to be replaced by another convex lens of same material having different radii of curvatures on both sides ($R_1 \neq R_2$), without any change in lens power then possible combination of R_1 and R_2 is :

(1) $\frac{1}{3}$cm and $\frac{1}{3}$cm

(2) $\frac{1}{5}$ cm and $\frac{1}{7}$ cm

(3) $\frac{1}{3}$cm and $\frac{1}{7}$ cm

(4) $\frac{1}{6}$ cm and $\frac{1}{9}$ cm

24) If $y = \sqrt{\sin\ \sqrt{x}}$ then $\frac{dy}{dx}$ is :

(1) $\sqrt{\cos\ \sqrt{x}}$

(2) $\frac{\sqrt{\cos\ \sqrt{x}}}{4\sqrt{x}}$

(3) $\frac{\cos\ \sqrt{x}}{4\sqrt{x\ \sin\ \sqrt{x}}}$

(4) $\frac{\cos\ \sqrt{x}}{4\sqrt{\sin\ \sqrt{x}}}$

25) **Assertion :** The electric and magnetic fields in electromagnetic waves are perpendicular to each other and to the direction of propagation.
Reason : Speeds of different electromagnetic waves are different in vacuum.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Both assertion and reason are false.

26) A locomotive of mass m starts moving so that its velocity varies according to the law $v = k\sqrt{S}$ where k is constant and S is the distance covered. Find the total work performed by all the forces which are acting on the locomotive during the first t seconds after the beginning of motion.

(1) $W = \frac{1}{8}\ mk^4t^2$

(2) $W = \frac{1}{4} m^2k^4t^2$

(3) $W = \frac{1}{4}mk^4t^4$

(4) $W = \frac{1}{8} mk^4t^4$

27) In relativistic theory, mass $m = \frac{m_0}{\sqrt{\left(1-\frac{v^2}{c^2}\right)}}$
The equivalent relation in electricity for electric charge is:-

(1) $q = q_0$

(2) $q = \frac{q_0}{\sqrt{\left(1-\frac{v^2}{c^2}\right)}}$

(3) $q_0 = \frac{q}{\sqrt{\left(1-\frac{v^2}{c^2}\right)}}$

(4) $q = \frac{q_0 v}{c}$

28) Two resistors of resistances $R_1 = (100 \pm 3)\Omega$ and $R_2 = (200 \pm 4)\Omega$ are connected in parallel. The equivalent resistance in parallel combination is:

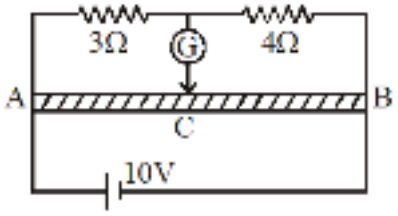
(1) $(66.7 \pm 1.8)\Omega$

(2) $(66.7 \pm 4.0)\Omega$

(3) $(66.7 \pm 3.0)\Omega$

(4) $(66.7 \pm 7.0)\Omega$

29) The galvanometer has no deflection and the length of wire AB is 350 cm. then AC equals to :-



(1) 150 cm.

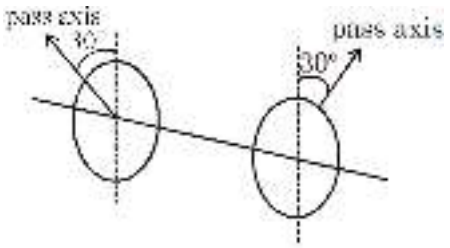
(2) 200 cm.

(3) 210 cm.

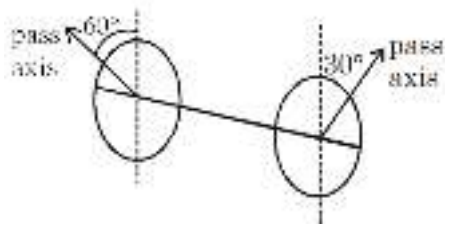
(4) 280 cm.

30) Unpolarized light of equal intensity is incident on four pairs of polarizing filters. Rank in order, from largest to smallest, the intensities I_a to I_d transmitted through the second polarizer of each pair :-

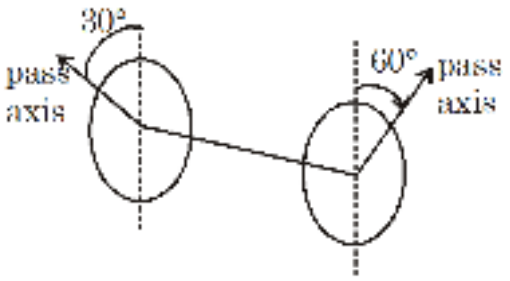
(a)



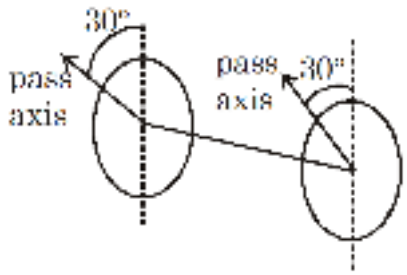
(b)



(c)



(d)



(1) a > b = c > d

(2) d > a > b = c

(3) b > c > a > d

(4) a > b > c > d

31) One mole of an ideal diatomic gas is expanded. During expansion, volume and temperature of gas vary such that $\frac{V}{T^2}$ = constant. Find heat transfer during expansion if temperature of the gas increases by 50K.

(1) 225R

(2) 125R

(3) 100R

(4) None of the above

32)

In the displacement method, a convex lens is placed in between an object and a screen. If the magnifications in the two positions are m_1 and m_2 and the displacement of the lens between the two positions is x, then the focal length of the lens is-

(1) $\frac{x}{m_1 + m_2}$

(2) $\frac{x}{m_1 - m_2}$

(3) $\frac{x}{(m_1 + m_2)^2}$

(4) $\frac{x}{(m_1 - m_2)^2}$

33) Electromagnetic waves are transverse in nature, is evident by

(1) polarization

(2) interference

(3) reflection

(4) diffraction

34) A metal ball of specific gravity 4.5 and specific heat 0.1 cal/gm-°C is placed on a large slab of ice at 0°C. When ball's temperature become 0°C then half of the ball sinks in the ice. The initial temperature (in °C) of the ball is:
(Latent heat of ice = 80 cal/g, specific gravity of ice = 0.9)

(1) 20

(2) 40

(3) 80

(4) 60

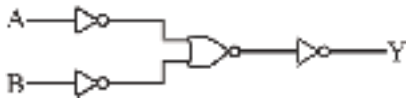
35) A wheel is spinning at 27 rad/s but is slowing with an angular acceleration that has a magnitude given by ($3t^2$ rad/s^2). It stops in a time of :-

(1) 1.7 s

(2) 2.6 s

(3) 3.0 s

(4) 4.4 s

36) Given combination of logic gates is equivalent to 

(1) AND gate

(2) XOR gate

(3) NOR gate

(4) NAND gate

37) When a glass capillary tube is immersed in a liquid, the liquid rises to a height of 6mm. The tube is gradually pressed down, until a length of only 4 mm projects outside. The liquid meniscus makes an angle of 60° with the walls of the capillary. Determine the angle of contact for the glass and the liquid pair.

(1) $\cos^{-1}(3/4)$

(2) $\cos^{-1}(3/7)$

(3) 37°

(4) $\cos^{-1}(1/2)$

38) When a mass M is attached to the spring of force constant k, then the spring stretches by □. If the mass oscillates with amplitude □, what will be the maximum potential energy stored in the spring?

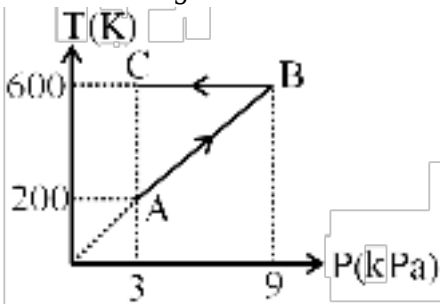
(1) $\frac{k\ell}{2}$

(2) 2k□

(3) $\frac{1}{2}$Mg□

(4) 2Mg□

39) 2 moles of a gas is taken from state A to C as shown in given T-P diagram then work done will be.



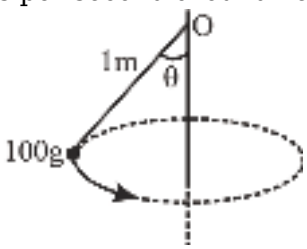
(1) 1200 R (□n 3)

(2) 900 R (□n 3)

(3) $\left(\ell n\frac{1}{3}\right)$ 400 R

(4) $\left(\ell n\frac{1}{3}\right)$ 1200 R

40) A string of length 1m is fixed at one end and carries a mass of 100 g at the other end. The string makes (2/π) revolutions per second around vertical axis passing through the fixed end. What is the tension in the string :-

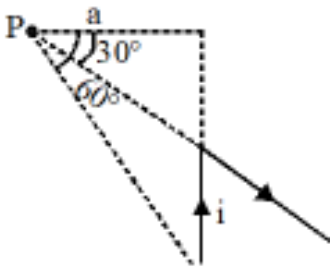


(1) 1.6 N

(2) 0.8 N

(3) 3.2 N

(4) 2.4 N



41) Find magnetic field at point P in given figure :-

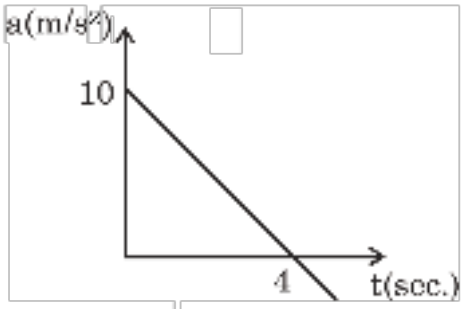
(1) $\frac{\mu_0 i}{4\pi a}\left(\sqrt{3}-1\right)$

(2) $\frac{\mu_0 i}{8\pi a}\left(1+\sqrt{3}\right)$

(3) $\frac{\mu_0 i}{8\pi a}\left(\sqrt{3}-1\right)$

(4) $\frac{\mu_0 i}{4\pi a}\left(1+\sqrt{3}\right)$

42) The acceleration-time graph of a particle moving along a straight line is as shown in figure. At what time the particle again acquires its initial velocity ?



(1) 12 sec

(2) 5 sec

(3) 8 sec

(4) 16 sec.

43) The resistance of wire is 5Ω at 50°C & 6Ω at 100°C. The resistance of the wire at 0°C will be :-

(1) 2Ω

(2) 1Ω

(3) 4Ω

(4) 3Ω

44) A student is performing the experiment of resonance column. The diameter of the column tube is 6 cm. The frequency of the tunning fork is 504 Hz. Speed of the sound at the given temperature is 336 m/s. The zero of the meter scale coincides with the top end of the resonance column tube. The reading of the water level in the column when the first resonance occurs is :-

(1) 13 cm

(2) 16.6 cm

(3) 18.4 cm

(4) 14.8 cm

45) Potential energy of a 3 kg body at the surface of a planet is –54 J, then escape velocity from the surface of planet will be :-

(1) 18 m/s

(2) 162 m/s

(3) 36 m/s

(4) 6 m/s

CHEMISTRY

1) If 150 calories of heat are added to a system and a work of 200 Joule is done on the system then calculate change in internal energy of the system.

(1) 430 Joule

(2) 830 Joule

(3) –830 Joule

(4) –430 Joule

2) Which of the following is incorrect ?

(1) Lycine, Arginine and Histidine are basic amino acid

(2) Deficiency of vitamin B_2 causes cheilosis

(3) Deficiency of vitamin B_6 causes convulsions

(4) Deficiency of vitamin B_2 causes Beri, beri

3) Redox titration of 20 ml of 0.5 M ferrous ammonium sulphate (Mohr's salt) against M/25 $KMnO_4$ solution is carried out in acidic medium. Calculate used volume of $KMnO_4$ at equivalence point or end point ?

(1) 25 ml

(2) 75 ml

(3) 80 ml

(4) 50 ml

4) The following data were obtained during the first order thermal decomposition of SO_2Cl_2 in closed container :-

$SO_2Cl_{2(g)} \rightarrow SO_{2(g)} + Cl_{2(g)}$

Experiment No.	**Time (in sec.)**	**Total pressure (in atm.)**
1	0	0.5
2	50	0.6

Calculate the rate constant :-

(1) $2.3 \times 10^{-3}\ s^{-1}$

(2) $4.6 \times 10^{-3}\ s^{-1}$

(3) $4.6 \times 10^{-4}\ s^{-1}$

(4) None of these

5) The rate of a first order reaction is 0.5 Ms^{-1} at 20 sec. and 0.4 Ms^{-1} at 30 sec. after initiation of the reaction. The half life time of the reaction is (approx.) :-

(1) 24 sec.

(2) 15 sec.

(3) 60 sec.

(4) 30 sec.

6) 6.7 gm Vanadium oxide (VO) reacts with 4.8 gm Fe_2O_3. Calculate amount of FeO, which is produced from following reaction.
[At. wt. of V = 51, Fe = 56]
$2VO + 3Fe_2O_3 \rightarrow 6FeO + V_2O_5$

(1) 4.32 gm

(2) 2.16 gm

(3) 6.22 gm

(4) 1.23 gm

7) An emission transition starts from the orbit having radius 1322.5 pm (picometer) and ends at 211.6 pm (picometer). Name the series of hydrogen spectrum to which this transition belongs :-

(1) Lyman series

(2) Balmer series

(3) Paschen series

(4) Pfund series

8) $CH_3COOH \xrightarrow{SOCl_2} A \xrightarrow[\Delta]{NH_3} B \xrightarrow[KOH]{Br_2} C \xrightarrow[KOH]{CHCl_3} D$
Identity D in above reaction sequence.

(1) CH_3CH_2NC

(2) CH_3–NC

(3) CH_3CH_2OH

(4) CH_3CN

9) Calculate pH of resultant mixture at 25°C when 500 ml of 0.1 M NH_4OH solution is treated with 250 ml of 0.1 M HCl. Given K_b of $NH_4OH = 2 \times 10^{-5}$:-

(1) 9.3

(2) 4.7

(3) 8.5

(4) 10.3

10) The metal which in +1 oxidation state in aqueous solution undergoes disproportionation is :

(1) Zn

(2) Mn

(3) Cu

(4) Cr

11) Suppose a gaseous mixture of F, Cl, Br and I is irradiated with photons of frequency appropriate to ionise Cl, then which ion (s) will be present in the mixture ?

(1) F^+ only

(2) F^+, Cl^+, Br^+ only

(3) Cl^+ only

(4) Cl^+, Br^+ and I^+ only

12) **Assertion (A) :** For the separation of group II cations H_2S gas is passed in acidic medium.
Reason (R) : In acidic medium due to common ion effect, concentration of S^{-2} is decreased to avoid precipitation of group IV cations.
Choose the correct option among following :

(1) Assertion is true but Reason is false.

(2) Assertion is false but Reason is true.

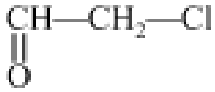
(3) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

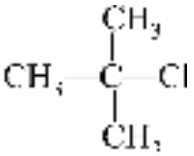
(4) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

13) Which of the following is most reactive towards S_N2 mechanism?

(1) CH_3–Cl

(2) $CH_2 = CH–CH_2–Cl$

(3) 

(4) 

14) The correct statement for carbonyl compound in follow are :
(A) Lower member of aldehyde and ketone are water soluble because they form H-bond with H_2O
(B) Boiling point of carbonyl compound lies between the boiling point of their correspondent ether & alcohol
(C) Racemic mixture is formed while new chiral centre is generated during nucleophilic addition reaction of carbonyl compound.
(D) Carboxylic acid are lower boiling liquid than aldehyde.

(1) Only A, B and C

(2) Only A and C

(3) Only A, B and D

(4) A, B, C and D

15) Which of the following conversion is correct with suitable reagent ?

	Reactions	**Reagents**
(1)	Benzophenone → Diphenyl methane	$LiAlH_4$
(2)	Benzaldehyde → 1-Phenyl ethanol	Zn(Hg)/ Conc HCl
(3)	Cyclohexanone → Cyclohexanol	CH_3MgBr
(4)	Phenyl benzoate → Benzaldehyde	DIBAL-H

(1) 1

(2) 2

(3) 3

(4) 4

16) The half life of a first order reaction is 6.93×10^{-3} min at 27°C. If at this temperature, value of pre-exponential factor A is 10^{12} min^{-1} then calculate the fraction of molecules of reactants having energy equal to or greater than activation energy.

(1) 10^{-8}

(2) 10^{10}

(3) 10^{-10}

(4) 10^{8}

17) One mole of P_2H_4 loses ten moles of electrons to form a new compound Y. Assuming that all the phosphorous atom appears in the new compound, what is the oxidation state of phosphorous in Y ?

(1) –1

(2) –3

(3) +3

(4) +5

18) For reactions :
(1) $A_{(g)} \rightleftharpoons B_{(g)} + C_{(g)}$ (D.O.D. = α_1)
(2) $X_{(g)} \rightleftharpoons 2Y_{(g)}$ (D.O.D. = α_2)
Given :-
$\frac{K_{P_1}}{K_{P_2}} = \frac{4}{3}$ and $\alpha_1 = 2\alpha_2$ (α_1 and $\alpha_2 << 1$)
Find value of $\left(\frac{P_1}{P_2}\right)$, where P_1 and P_2 are total pressure at equilibrium for 1st and 2nd reaction respectively.

(1) $\frac{16}{3}$

(2) $\frac{4}{12}$

(3) $\frac{8}{3}$

(4) $\frac{4}{3}$

19) From the given set of species, identify the species from each set having minimum radius :
(A) O^{-2}, F^{-}, Na^{+} (B) Ni, Cu, Zn
(C) Li, Be, Mg (D) Sc, Y, La

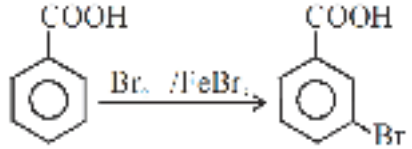
(1) O^{-2}, Cu, Li

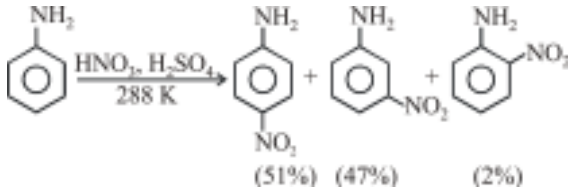
(2) Na^{+}, Ni, Be, Sc

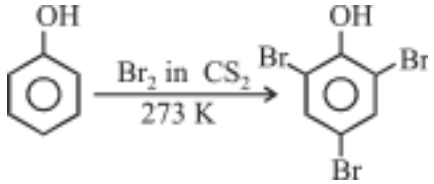
(3) F^{-}, Zn, Mg, Y

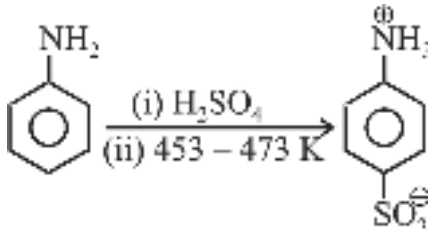
(4) Na^{+}, Zn, Be, La

20) Which of the following reaction is incorrect ?

(1) 

(2) 

(3) 

(4) 

21) Choose the incorrect statement from the following :

(1) In $N_2 \rightarrow N_2^{+}$, the electron is lost from $\sigma 2p_z$ orbital

(2) In $O_2 \rightarrow O_2^{+}$, the electron is lost from $\pi^* 2p_x$ or $\pi^* 2p_y$ orbital

(3) In $O_2 \rightarrow O_2^{-}$, the electron is added into $\pi^* 2p_x$ or $\pi^* 2p_y$ orbital

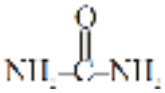
(4) In $N_2 \rightarrow N_2^{-}$, the electron is added into $\sigma^* 2p_z$ orbital

22) Kjeldahl's method cannot be used to estimate nitrogen for which of the following compounds?

(1) $C_6H_5NO_2$

(2) $C_6H_5NH_2$

(3) CH_3CH_2–C≡N

(4) 

23) A conductivity cell contains 0.1 M HCl solution, which have 75Ω resistance and $4 \times 10^{-2}\ \Omega^{-1}\ cm^{-1}$ conductivity. If this cell is filled with 0.05 M solution of unknown electrolyte which have 250 Ω resistance then calculate molar conductivity of 0.05 M solution of unknown electrolyte ?

(1) 240 $\Omega^{-1}.cm^2.mol^{-1}$.

(2) 120 $\Omega^{-1}.cm^2.mol^{-1}$.

(3) 360 $\Omega^{-1}.cm^2.mol^{-1}$.

(4) 480 $\Omega^{-1}.cm^2.mol^{-1}$.

24) **Assertion (A) :** In some of the reactions T□ resembles Al, whereas in others it resembles with alkali metals.
Reason (R) : T□ shows both +1 or +3 oxidation states in its compounds.

(1) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

25) ***Assertion :-*** Acetone and Acetaldehyde can be distinguished by Tollen's reagent.
Reason :- Aldehyde give silver mirror with Tollen's reagent but Ketone does not.

(1) Assertion is true reason is true and reason is correct explanation of assertion

(2) Assertion is true reason is true but not correct explanation of assertion.

(3) Assertion is true reason is false

(4) Assertions is false reason is true.

26) Which of the following order of conductivity in aqueous solution for following complexes is correct ?
(I) $CoCl_3.6NH_3$ (II) $CoCl_3.5NH_3$
(III) $CoCl_3.4NH_3$

(1) III > II > I

(2) I > II > III

(3) III > I > II

(4) I > III > II

27) Conc. HNO_3 can be transported in container of :

(1) Zn

(2) Cu

(3) Ni

(4) Al

28) Find E_{cell} for the following cell :
$Mg_{(s)}|Mg^{+2}(0.10\ M)||Ag^{+}(0.001\ M)|Ag_{(s)}$
Given : $E^0_{cell} = 3.17\,V$
$\frac{2.303RT}{F} = 0.06$

(1) 3.17 V
(2) 3.02 V
(3) 3.32 V
(4) 2.86 V

29) Aqueous solution of Ni^{2+} contains $[Ni(H_2O)_6]^{2+}$ and its magnetic moment is 2.83 B.M. When ethylene di amine is added into excess then what will be spin magnetic moment of formed complex ?

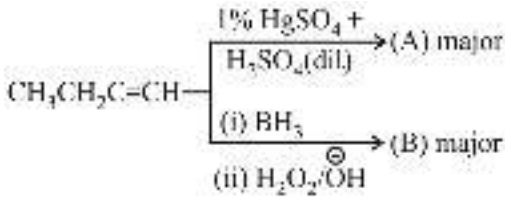
(1) It will remain same
(2) It increases from 2.83 B.M.
(3) It decreases from 2.83 B.M.
(4) It cannot be predicted theoretically

30) At same temperature 0.6% w/v solution of urea would be isotonic with :-

(1) 0.1 M glucose solution
(2) 0.1 M NaCl solution
(3) 0.9% w/v glucose solution
(4) 7.45% w/v KCl solution

31) Which of the following complex is likely to show optical activity ?

(1) Trans-$[CoCl_2(NH_3)_4]^+$
(2) $[Cr(H_2O)_6]^{3+}$
(3) Cis-$[Co(NH_3)_2(en)_2]^{3+}$
(4) Trans-$[Co(NH_3)_2(en)_2]^{3+}$

32)


Relationship between product (A) and (B) :-

(1) Position isomers
(2) Diastereomers
(3) Geometrical isomers
(4) Functional group isomers

33) The enthalpy changes for two reactions are given by the equations :

$$2Cr_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Cr_2O_{3(s)};\ \Delta H = -1130 kJ$$

$$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)};\ \Delta H = -110 kJ$$

What is the enthalpy change (in kJ) for the following reaction?

$3C_{(s)} + Cr_2O_{3(s)} \rightarrow 2Cr_{(s)} + 3CO_{(g)}$

(1) –1460 kJ

(2) –800 kJ

(3) +800 kJ

(4) None of these

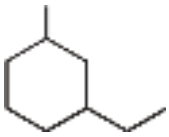
34) In which is the following pairs both species are iso structural ?

(1) NO_2^+ and NO_2^-

(2) PCl_5 and BrF_5

(3) ClO_3^- and SO_3^{-2}

(4) SeF_4 and XeO_4

35) The IUPAC name of given compound is :- 

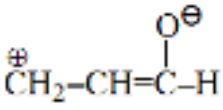
(1) 1-Ethyl-3-methyl cyclohexane

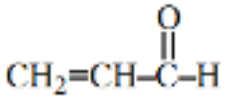
(2) 1-Methyl-3-ethyl cyclohexane

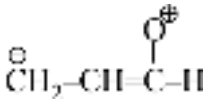
(3) 3-Methyl-1-ethyl cyclohexane

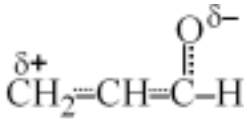
(4) 3-Ethyl-1-methyl cyclohexane

36) Acryl aldehyde have three resonating structure (I to III) and a resonating hybrid (IV) which are given below :-

(I) 

(II) 

(III) 

(IV) 

Resonance stabilisation energy is the difference of :-

(1) I and II

(2) I and III

(3) I and IV

(4) II and IV

37) In which of the following molecule all fluorine atoms are in same plane ?

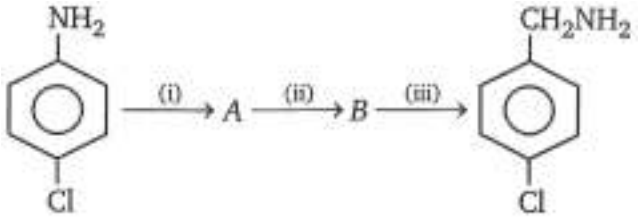
(1) XeF_2

(2) ClF_3

(3) XeF_4

(4) All of these

38) Mark the correct route of the conversion of p-chloroaniline to p-chlorobenzylamine.



	i	ii	iii
(1)	Alkylation	KCN	H_2/Pt
(2)	Diazotisation	CuCN	H_2/Pt
(3)	Oxidation	H_2/Pt	Hydrolysis
(4)	Diazotisation	H_2O/H^+	Sn/HCl

(1) 1

(2) 2

(3) 3

(4) 4

39) Consider following statements :
Statement-I : The enthalpy of atomisation is an important factor in determining the standard electrode potential of a metal.
Statement-II : Metals with very high enthalpy of atomisation tend to be noble in their reactions.
Then which option is correct ?

(1) Only statement-I is correct.

(2) Only statement-II is correct.

(3) Both statement-I and statement-II are correct.

(4) Both statement-I and statement-II are incorrect.

40) 3 molal NaOH solution has a density 1.11 g/ml. Find molarity of this solution.

(1) 2.05 M

(2) 2.97 M

(3) 2.45 M

(4) 3.21 M

41) Incorrect statement among following is -

(1) Electromeric effect is a temporary effect

(2) In electromeric effect π-bonded electron are completely transfer in presence of reagent

(3) In electromeric effect heterolytic bond breaking takes place in σ bond

(4) When inductive and electromeric effect operates in opposite directions the electromeric effect predominates

42) Which of the following order is not correct for hydrides of group 15 elements ?

(1) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$; Thermal stability

(2) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$; Reducing nature

(3) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$; Basic strength

(4) $BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$; Boiling point

43) The species that will have the maximum dipole moment is :

(1) CH_3Cl

(2) CH_2Cl_2

(3) $CHCl_3$

(4) CH_3F

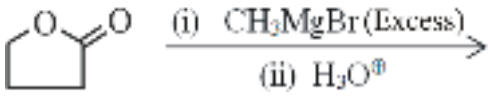
44) ***Assertion :-*** t-butyl methyl ether is prepared by reaction of t-butyl bromide with methoxide ion.
Reason :- The above reaction follow S_N2-mechanism.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

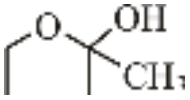
(4) Both Assertion & Reason are False.

45) 

X,
product X is

(1) 

(2) 

(3) 

(4) 

BIOLOGY

1) Which of the following statement is not correct ?

(1) A number of traditional drinks and foods are made by fermentation by microbes.

(2) Brandy and rum are produced without distillation of the fermented broth.

(3) Antibiotics are 'pro life' with reference to human beings.

(4) Statins competitively inhibit the enzyme responsible for synthesis of cholesterol.

2) Which of the following statement is **not correct** ?

(1) A stable community should not show too much variation in productivity from year to year.

(2) The great german naturalist and geographer Alexander von Humboldt observed that within a region species richness increases with increasing explored area, but only up to a limit.

(3) Temperate environments, unlike tropical ones, are less seasonal, relatively more constant and predictable.

(4) The biological wealth of our planet has been declining rapidly and the accusing finger is clearly pointing to human activities.

3) Given below are two statements :-
Statement-I : The stage between the two meiotic divisions is called interkinesis and is generally short lived, there is no replication of DNA during this phase and it is followed by prophase II a much simpler than prophase I.
Statement-II : Meiosis also increases the genetic variability in the population of organisms from one generation to the next and variations are very important for the process of evolution.
In the light of above statements, choose the correct answer from option given below :-

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

4) During a recombinant DNA experiment a researcher observed that the vector used showed blue-white screening and contained multiple restriction site within a reporter gene
Statement-I :- The vector allows identification of recombinants due to insertional inactivation.
Statement-II :- The reporter gene involved in lac-Z, which code for β-galactosidase
Choose the correct option :-

(1) Both statement are true

(2) Both statement are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement II is correct

5) Given below are two statements:
Statement I : In eukaryotes there are at least three RNA polymerases in the nucleus (in addition to the RNA polymerase found in the semi-autonomous organelles).
Statement II : All the three RNA polymerases in eukaryotic nucleus have different roles.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) **Statement I** is correct but **Statement II** is incorrect

(2) **Statement I** is incorrect but **Statement II** is correct

(3) Both **Statement I** and **Statement II** are correct

(4) Both **Statement I** and **Statement II** are incorrect

6) Arrange the following events of meiosis-I in correct sequence.
(a) Crossing over
(b) Synapsis
(c) Chiasmata
(d) Nucleolus disappears

(1) a, b, c, d

(2) c, b, a, d

(3) b, a, c, d

(4) d, c, b, a

7) Read the following statement :-
(A) ADA disorder is caused due to the deletion of the gene for adenosin deaminase.
(B) RNAi takes place in all eukaryotic organisms as a method of cellular defense.
(C) Bt forms protein crystals contain a toxic insecticidal protein.
(D) GMO helped to reduce post harvest losses.
Choose the correct option.

(1) A, B

(2) A, B, C

(3) B, C, D

(4) A, B, C and D

8) Identify the correct match from column I,II and III :-

	Column-I		**Column-II**		**Column-III**
I	Replication	a	RNA→DNA	i	Ribosome
II	Transcription	b	DNA→DNA	ii	RNA-dependent DNA polymerase
III	Translation	c	DNA→RNA	iii	DNA-dependent DNA polymerase
IV	Reverse Transcription	d	mRNA→Polypeptide	iv	DNA-dependent RNA polymerase

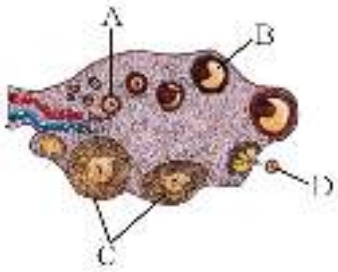
Options :-

(1) I-a-ii, II-b-iii, III-c-iv, IV-d-i

(2) I-b-iii, II-c-iv, III-d-i, IV-a-ii

(3) I-d-i, II-a-ii, III-b-iii, IV-c-iv

(4) I-c-iv, II-d-i, III-a-ii, IV-b-iii



9) Identify the correct one from given diagram :

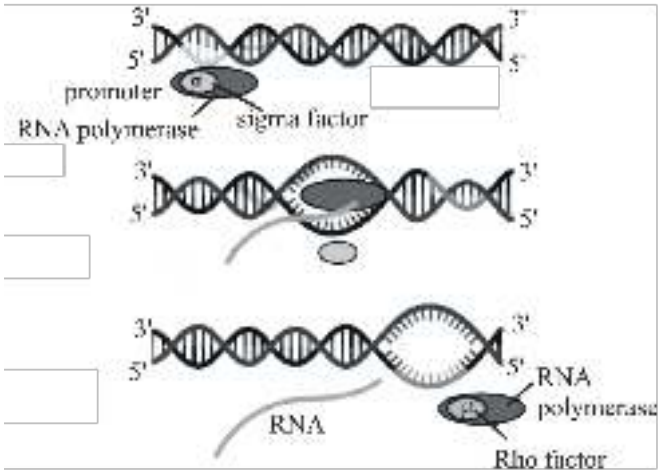
(1) A-Primary follicle-The number retains constant from birth to puberty
(2) B-Secondary follicle-Presence of antrum
(3) C-Corpus luteum-Secretes large amount of progesterone
(4) D-Secondary oocyte-Released during the beginning of the menstrual cycle

10) In the given table fill ART and Method in blanks-

ART	Methods
i	Zygote upto 8 cell stage transfer in to the fallopian tube
ii	Sperm directly injected in to ovum
GIFT	iii

(1) i-ZIFT, ii, ICSI, iii-Transfer of ovum into the oviduct.
(2) i-ICSI, ii, ZIFT, iii-Inseminate the sperm into vagina
(3) i-IUI, ii, GIFT, iii-Transfer of ovum into the vagina
(4) i-ZIFT, ii, IUI, iii-Transfer of sperm into vagina

11) Given below diagram represents which of the following process ?



(1) Replication
(2) Transcription in Eukaryotes
(3) Transcription in Prokaryotes
(4) Splicing

12) Identify the wrong statements :-
(A) Algae reproduce only sexually.
(B) Liverwort have upright slender axes bearing spirally arranged leaves.

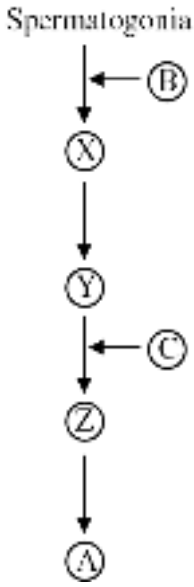
(C) In pteridophytes, the gametophytes bear male and female sex organs called antheridia and archegonia, respectively.
(D) Ovules in gymnosperm are not enclosed by an ovary wall.
Choose the most appropriate answer from the options given below :-

(1) B and D only

(2) B and C only

(3) A and B only

(4) A and C only



13) above is schematic representation of spermatogenesis
Find the option which identify (X) (Y) (Z) (B) (C) (A)

	X	Y	Z	B	C	A
1	Secondary spermatocyte	Spermatid	Sperm	Ist meiotic division	differentiation	Primary spermatocyte
2	Secondary spermatocyte	Sperm	2nd meiotic division	Ist meiotic division	differentiation	Primary spermatocyte
3	Primary spermatocyte	Secondary spermatocyte	Spermatid	Mitotis and differentiation	2nd meiotic division	Sperm
4	Primary spermatocyte	Sperm	Sperm	Mitotic differentiation	2nd meiotic division	Spermatid

(1) 1

(2) 2

(3) 3

(4) 4

14) Three type of areas ie sensory areas, motor areas & association areas are found in which part of brain?

(1) Cerebellum

(2) Medulla oblongata

(3) Cerebral cortex

(4) Midbrain

15) Which of the following characteristic represent inheritance of blood group in human ?
(i) ABO blood groups are controlled by the gene I.
(ii) Plasma membrane of the red blood cells has amino-polymers that protrude from its surface
(iii) The gene I has three alleles i.e. I^A, I^B and i. Each diploid person possesses all three alleles of gene I.
(iv) There are eight different possible combinations of three alleles of gene I.
(v) 'O' blood group is not possible in the progeny of 'AB' blood group parents.

(1) only i and v

(2) only ii, iii and iv

(3) only ii and iv

(4) only i, iii and v

16) Which of the following statement is **wrong** ?

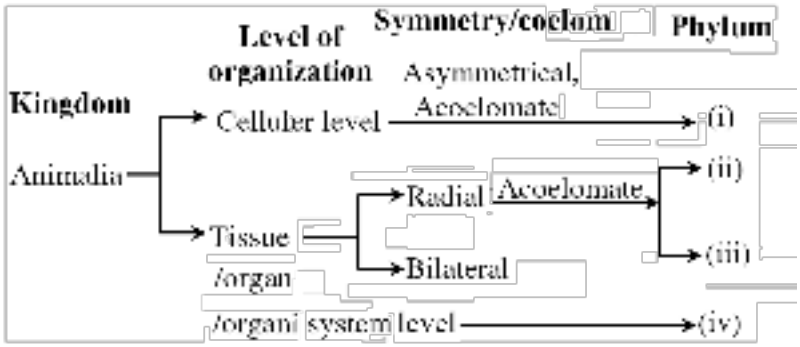
(1) Rate of photosynthesis can be directly proportional to CO_2 concentration upto a limit

(2) Rate of photosynthesis can be directly proportional to the number of leaves

(3) Usually light is not a limitting factor for photosynthesis

(4) Plant or internal factors affecting photosynthesis depend only on plant growth hormones.

17) Observe the given flow chart carefully and fill the boxes with appropriate words and select the correct option.



	(i)	(ii)	(iii)	(iv)
1	Porifera	Coelentrerata	Ctenophora	Arthropoda
2	Porifera	Mollusca	Coelenterata	Annelida
3	Arthropoda	Porifera	Coelenterata	Mollusca
4	Mollusca	Arthropoda	Annelida	Echinodermeta

(1) 1

(2) 2

(3) 3

(4) 4

18) **Assertion :-** Thyroid hormone play an important role in the regulation of the basal metabolic rate. These hormones also support the process of red blood cell formation.
Reason :- Thyroid hormone control the metabolism of carbohydrate, proteins and fats. Maintenance of water and electrolyte balance is also influenced by thyroid hormones.

(1) Both Assertion and Reason are true and Reason is a correct explaination of Assertion

(2) Both Assertion and Reason are false

(3) Both Assertion and Reason are true and Reason is not a correct explaination of Assertion

(4) Assertion is true and Reason is false

19) The given table shows the order of evolutionary stages of humans. Choose the correct option to fill the blanks in the table given below.

Early Ancestor	**Later Ancestor**
_ _ i _ _	Australopithecus
_ _ ii _ _	Homo erectus
Cromagnon man	_ _ iii _ _

(1) i-Ramapithecus; ii-Neanderthal man; iii-Homo sapiens sapiens

(2) i-Homo habilis; ii-Java man; iii-Neanderthal man

(3) i-Homo habilis; ii-Peking man; iii-Neanderthal man

(4) i-Ramapithecus; ii-Homo habilis; iii-Homo sapiens sapiens

20) While studying a genetically modified crop plant a researcher observed that the plant shows resistance to specific insect pests without the application of chemical insecticide. Histological examination revealed damage to the midgut epithelium cells of insects feeding on the plant due to pore formation. What is the most likely reason for this resistance ?

(1) Production of active toxin by plant.

(2) Activation of plant defense system.

(3) RNA interference in insect gut cells.

(4) Bt-toxin is converted into an active form of toxin due to the alkaline pH of the gut of insect.

21) In plants like *Amorphophallus*, the flowers are pollinated by :

(1) Wind

(2) Insect

(3) Water

(4) Birds

22) Select **incorrect** statement :-

(1) Adventive embryony is found in citrus and mango.

(2) Technically megasporangium is equivalent to ovule.

(3) In cereals such as maize, cotyledon is called scutellum.

(4) Epiblast is found in dicotyledonous embryo.

23) The members of ascomycetes differ from basidiomycetes in having :

(1) Branched and septate mycelium
(2) Conidia formation
(3) Formation of haploid sexual spores
(4) Formation of diploid sexual spores

24) If in a pond there were 40 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 48, calculate the birth rate:

(1) 0.1 offspring per lotus per year.
(2) 0.2 offspring per lotus per year.
(3) 0.3 offspring per lotus per year.
(4) 0.4 offspring per lotus per year.

25) Seminal plasma in humans contains :

(1) Fructose, hyaluronidase enzymes and calcium
(2) Fructose, calcium, certain enzymes
(3) Glucose, proacrosin and hyaluronidase enzymes
(4) Glucose, proacrosin, calcium and phosphate

26) Following are the steps involved in generation of action potential :-
(A) Opening of Na^+ voltage-gated channel
(B) Reversal of membrane polarity
(C) Stimulus is applied on polarised membrane
(D) Rapid influx of Na^+ ions
(E) Generation of action potential
Choose the correct sequence of steps from the options given below :-

(1) C → D → A → B → E
(2) C → D → B → A → E
(3) C → A → D → B → E
(4) C → A → B → D → E

27) All animals depend on plants (directly or indirectly) for their food needs, they are hence called _____ and also heterotrophs.

(1) Herbivores
(2) Producers
(3) Consumers
(4) Carnivores.

28) Regarding electron transport system match the following columns.

	Column I		Column II
a.	Complex I	i.	Transfers electrons to cytochrome 'c'
b.	Complex II	ii.	Receives electrons from $FADH_2$
c.	Complex III	iii.	Pumps protons and transfers electrons from NADH
d.	Complex IV	iv.	Reduces oxygen to water

(1) a-iii, b-ii, c-i, d-iv

(2) a-ii, b-iii, c-iv, d-i

(3) a-iv, b-ii, c-iii, d-i

(4) a-iii, b-iv, c-i, d-ii

29) Select the incorrect option ?

(1) Pivot joint → Between atlas and axis

(2) Hinge joint → Knee joint

(3) Gliding joint → Between carpal and Metacarpal of thumb

(4) ball and Socket joint → Between humerus and pectoral girdle

30) An experiment visually demonstrates Kranz anatomy, a key adaptation in C-4 plants. The distinct arrangement of mesophyll and bundle sheath cells supports the separation of the light reaction and Calvin cycle, allowing efficient photosynthesis even under high light and temperature conditions.
In relation to above mentioned paragraph read the following statements :
Statement-I : In C-4 plants, the initial fixation of CO_2 occurs in the mesophyll cells, forming oxaloacetic acid, it then forms other 4-carbon compound which is later decarboxylated in the bundle sheath cells.
Statement-II : This anatomical adaptation allows C-4 plants to concentrate CO_2 in bundle sheath cells, thereby reducing photorespiration and enhancing photosynthetic efficiency, especially under high temperature and light intensity.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) **Statement I** is correct but **Statement II** is incorrect.

31) Match the **column-I** with **column-II** correctly.

Column - I [Category]		Column - II [Example(s)]	
A	Pigments	1	Concanavalin A
B	Terpenoides	2	Monoterpenes, Diterpenes
C	Alkaloids	3	Morphine, codeine
D	Lectins	4	Carotenoids, Anthocyanins

(1) A – 4, B – 2, C – 3, D – 1

(2) A – 4, B – 3, C – 2, D – 1

(3) A – 1, B – 4, C – 3, D – 2

(4) A – 1, B – 3, C – 2, D – 4

32) When a heterozygous tall pea plant of F_1 generation upon self fertilization produces tall and dwarf phenotypes it proves the principle of :-

(1) Incomplete Dominance

(2) Segregation

(3) Independent assortement

(4) Codominance

33) **Statement-I :** In fermentation there is a net gain of only two molecules of ATP for each molecules of glucose degradation whereas many more molecules of ATP are generated under aerobic conditions.
Statement-II : NADH is oxidised to NAD^+ rather slowly in fermentation, however the reaction is very vigorous in case of aerobic respiration.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

34) Read the following statements about ecological pyramids and identify the **incorrect** ones.
(a) Generally in terrestrial ecosystems, pyramids of number, biomass and energy are upright.
(b) In an aquatic ecosystem, pyramid of biomass is always upright.
(c) The apex of an ecological pyramid represents decomposers.
(d) Ecological pyramids do not represent saprophytes even though they play an important role in the ecosystem.
(e) Pyramid of energy can never be inverted because energy is lost as heat at each trophic level.

(1) b and c only

(2) a, b and c only

(3) b, c and d only

(4) b and d only

35) Sclerenchymatous hypodermis, sclerenchymatous bundle sheath and conjoint closed vascular bundles are found in :-

(1) Dicot root

(2) Monocot root

(3) Dicot stem

(4) Monocot stem

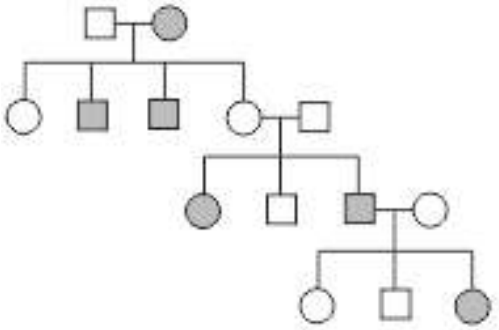
36) Leaf develops :-

(1) Endogenously at the node

(2) Exogenously from internode

(3) Endogenously from internode

(4) Exogenously at the node



37) Below given pedigree chart show inheritance of -

(1) Sickle cell Anemia

(2) Haemophillia

(3) Colourblindness

(4) Myotonic dystrophy

38) Given below are two statements:
Statement-I: Homeotherm, skin is dry without gland except oil gland, Bones are hollow with air cavity, Air sac connected to the lungs, digestive tract has additional chamber crops, gizzard. Member of this class included in Aves.
Statement-II: Blue whales live in water but respire through lungs and whale are viviparous animal.
In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both statement-I and statement-II are incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Statement-I is correct but statement-II is incorrect

(4) Both statement-I and statement-II are correct

39) Match the column-A with column-B and select correct option :-

	Column-A		**Column-B**
A.	Engulf food by pseudopodia	i.	Ciliated protozoan
B.	Cause sleeping sickness	ii.	*Amoeba*
C.	Most notorious to humans	iii.	*Plasmodium*
D.	Have gullet	iv.	Flagellated protozoan

(1) A-ii, B-iii, C-i, D-iv

(2) A-iv, B-i, C-iv, D-iii

(3) A-ii, B-iv, C-iii, D-i

(4) A-ii, B-iv, C-i, D-iii

40) **Assertion:** Plants need the help of animals for pollinating their flowers and dispersing their seeds.
Reason: Plants offer rewards in the from of pollen and nectar for pollinators and juicy and nutritious

fruits for seed dispersal.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(2) **Assertion** is true but **Reason** is false.
(3) **Assertion** is false but **Reason** is true.
(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

41) Which of the following set is of **natural** plant growth regulators ?

(1) Zeatin, 2,4-D, GA_3
(2) Kinetin, Ethylene, NAA
(3) IAA, Zeatin, GA_3
(4) Kinetin, ABA, NAA

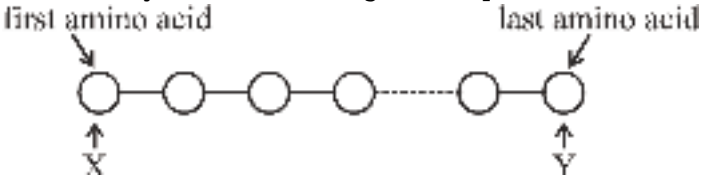
42) How much carbon dioxide is removed by lungs?

(1) 200 mL/second
(2) 200 mL/minute
(3) 200 mL/ day
(4) 200 mL/hour

43) The 9+2 arrangement of flagella refers to :-

(1) 9 microfilaments and 2 microtubules.
(2) 9 radial spokes and 2 central microtubules.
(3) 9 triplets of peripheral microtubules and 2 central microtubules
(4) 9 Doublet of peripheral microtubules and 2 central microtubules.

44) Identify X and Y in the given sequence of amino acids :-



	X	Y
1	N-terminal amino acid	C-terminal amino acid
2	N-terminal amino acid	N-terminal amino acid
3	C-terminal amino acid	N-terminal amino acid
4	C-terminal amino acid	C-terminal amino acid

(1) 1
(2) 2
(3) 3
(4) 4

45) Read all statement carefully and select **correct** statement from given option below :

(1) The trichome in the shoot system are usually unicellular.
(2) In grasses the guard cells are dumb-bell shaped.
(3) Phloem parenchyma is absent in most of dicot plants.
(4) The root hairs are unicellular elongation of the hypodermal cells and help absorb water from the soil.

46) Independent assortment of genes does not takes place when :–

(1) Genes are located on two different homologous chromosome pairs.
(2) Genes are tightly linked and located on same chromosome
(3) Genes are located on non–homologous chromosomes.
(4) All of the above

47) Which of the following statement is wrong ?

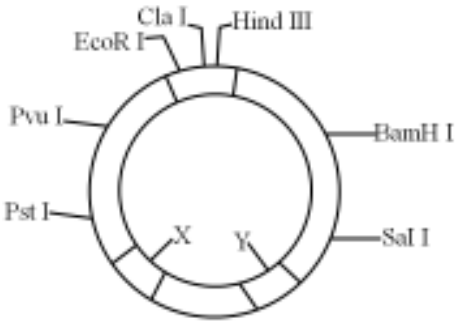
(1) Marginal placentation are found in Pea.
(2) False septum or replum are present in free central placentation.
(3) When the ovules are borne on central axis and septa are absent, it is called free central.
(4) Basal placentation is found in sunflower.

48) Match the following columns and select correct option.

	I		**II**		**III**		**IV**
P	Phycomycetes	i	Sac fungi	A	Conidia	W	*Puccinia*
Q	Ascomycetes	ii	Club fungi	B	Zoospores	X	*Rhizopus*
R	Basidiomycetes	iii	Imperfect fungi	C	Ascospores	Y	*Alternaria*
S	Deuteromycetes	iv	Lower fungi	D	Basidiospores	Z	*Neurospora*

(1) P-(iv)-A-Y ; Q-(i)-C-Z ; R-(ii)-D-W; S-(iii)-B-X
(2) P-(ii)-C-Z ; Q-(iii)-B-X ; R-(iv)-A-Y; S-(i)-D-W
(3) P-(iv)-B-X ; Q-(i)-C-Z ; R-(ii)-D-W; S-(iii)-A-Y
(4) P-(iii)-A-Y ; Q-(ii)-D-W ; R-(i)-C-Z; S-(iv)-B-X

49) The following diagram showing restriction site in E.coli cloning vector pBR 322. find the role of x & y gene.



(1) The gene x is responsible for resistance to antibiotics and 'y' for protein involved in replication of plasmid

(2) Gene x-responsible for controlling the copy no. of linked DNA,
Gene y-protein involved in replication

(3) Gene x-protein involved in replication,
Gene y-resistance to antibiotics

(4) Gene x-responsible for recognition sites,
Gene y-Antibiotic resistance

50) A bacterium containing 100% N^{15} N_2 bases is allowed to replicate in a medium containing N^{14} bases after one round of duplication the ratio of light : hybrid : heavy DNA will be

(1) 1 : 1 : 1

(2) 0 : 1 : 0

(3) 1 : 0 : 0

(4) 1 : 0 : 1

51) Match the column-I with column-II.

	Column-I		**Column-II**
(a)	Algae	(i)	*Cedrus*, *Pinus*
(b)	Liverwort	(ii)	*Ulothrix*, *Fucus*
(c)	Fern	(iii)	*Marchantia*, *Riccia*
(d)	Gymnosperm	(iv)	*Salvinia*, *Adiantum*

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iii), b-(i), c-(iv), d-(ii)

(3) a-(iii), b-(iv), c-(ii), d-(i)

(4) a-(ii), b-(iii), c-(iv), d-(i)

52) Identify the statement that is NOT correct.

(1) Antigen binding site is located at N-terminal region of antibody molecules.

(2) Each antibody has two light and two heavy chains.

(3) The heavy and light chains are not held together by disulfide bonds.

(4) The T-cells them selves do not secrete antibodies but help B-cells to produce them.

53) Which of the following structure does not open into the cloaca of a female frog?

(1) Rectum

(2) Oviduct

(3) Urinogenital duct

(4) Both 1 and 2

54) Open circulatory system is present in _____ and _____ in which blood pumped by the heart passes through large vessels into open spaces of body cavity called sinuses.

(1) Arthropoda, Molluscs

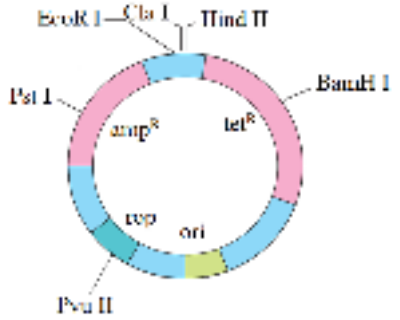
(2) Annelida, Hemichordate

(3) Reptilia, Mammalia

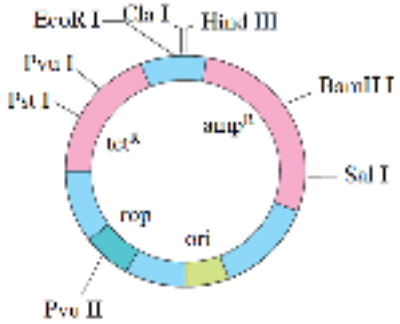
(4) Echinodermata, Chordata

55) Which of the following diagram is correct regarding to pBR322?

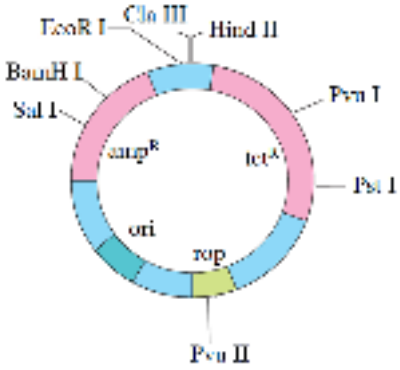
(1)



(2)



(3)



(4)



56) **Assertion (A) :-** The proton gradient across the thylakoid membrane is crucial for ATP synthesis.
Reason (R) :- The movement of proton down their gradient through the F_0-part of ATP synthase enzyme provides energy for phosphorylation.

(1) Both A & R are true and R is the correct explanation of A.

(2) Both A & R are true but R is not the correct explanation of A.

(3) A is true, R is false.

(4) Both A & R are false.

57) **Assertion (A) :** The continued oxidation of respiratory intermediates via the TCA cycle requires the continued regeneration of oxaloacetic acid and also requires regeneration of NAD^+ and FAD from NADH and $FADH_2$ respectively.
Reason (R) : The TCA cycle starts with the condensation of succinyl group with oxaloacetic acid and water and at few points in the cycle NAD^+ and FAD are reduced to NADH and $FADH_2$ respectively.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

58) Identify the meiotic stage in which homologous chromosome separate but sister chromatid remain associated :

(1) Anaphase I

(2) Anaphase II

(3) Metaphase I

(4) Metaphase II

59) The hypothalamus regulates the anterior pitutary's hormone secretion through a hypophyseal portal system. Which of the following hormones, secreted by the anterior pituitary, are regulated by the releasing hormones from the hypothalmus?

(1) GH & Prolactin

(2) TSH & ACTH

(3) Both 1 & 2

(4) Oxytocin & ADH

60) Match List-I with List-II.

	List-I		**List-II**
(A)	Colour-blindness	(I)	Polygenic inheritance
(B)	Phenylketonuria	(II)	Autosomal dominant
(C)	Myotonic dystrophy	(III)	Autosomal recessive
(D)	Human skin colour	(IV)	Sex linked recessive

Choose the correct answer from the options given below :-

(1) (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

(2) (A)-(II), (B)-(IV), (C)-(III), (D)-(I)

(3) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

(4) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

61)

	Column-I		**Column-II**
(A)	Lactic acid bacteria	(i)	Rise of bread dough
(B)	*Sacchromyces cerevisiae*	(ii)	Production of curd
(C)	*Penicillium notatum*	(iii)	Biogas production
(D)	Methanogenes	(iv)	Antibiotic production

(1) A-ii, B-iv, C-i, D-iii

(2) A-ii, B-i, C-iv, D-iii

(3) A-iii, B-ii, C-i, D-iv

(4) A-iv, B-i, C-ii, D-iii

62) Match the **Column I** and **Column II**

	Column I		**Column II**
(i)	Tidal volume (TV)	(a)	1100-1200 ml
(ii)	Inspiratory reserve volume	(b)	4800 ml
(iii)	Residual volume (RV)	(c)	500 ml
(iv)	Vital capacity	(d)	2500–3000 ml

Choose appropriate option -

(1) i-a, ii-b, iii-c, iv-d

(2) i-c, ii-d, iii-a, iv-b

(3) i-c, ii-b, iii-a, iv-d

(4) i-a, ii-c, iii-d, iv-b

63) Which of the following statement is **correct** about apomixis ?

(1) Apomixis is a type of sexual reproduction.

(2) Apomixis can help to maintain hybrid vigour across generations.

(3) Apomixis leads to the formation of genetically variable seeds.

(4) Apomixis involves meiosis and fertilisation.

64) In cockroach, sclerites of dorsal side tergum, ventral side sternum and of lateral side are called pleurons. These chitin plates attached with flexible membrane called :

(1) Peritrophic membrane

(2) Articular membrane

(3) Both

(4) Nictatiting membrane

65) In _____ the feeding efficiency of one species might be reduced due to interfering and inhibitory presence of the other species

(1) Competitive release
(2) Interference competition
(3) Mutualism
(4) Commensalism

66) Epipetalous and polyadelphous conditions are found in respectively?

(1) Citrus and *Petunia*
(2) *Petunia* and *Citrus*
(3) Mustard and Pea
(4) Pea and *Citrus*

67) Read the following terminology of column I & II and identify correct match :

	Column-I		**Column-II**
(A)	Restriction endonuclease	(I)	Sea weeds
(B)	Agarose	(II)	Bacteria
(C)	Plasmid	(III)	E.coli cloning vector
(D)	pBR322	(IV)	Bacterial extra chromosomal-DNA
(E)	Ti plasmid	(V)	Agrobacterium tumifaciens

(1) A-III, B-IV, C-V, D-I, E-II
(2) A-IV, B-V, C-I, D-II, E-III
(3) A-II, B-I, C-IV, D-III, E-V
(4) A-I, B-IV, C-III, D-V, E-II

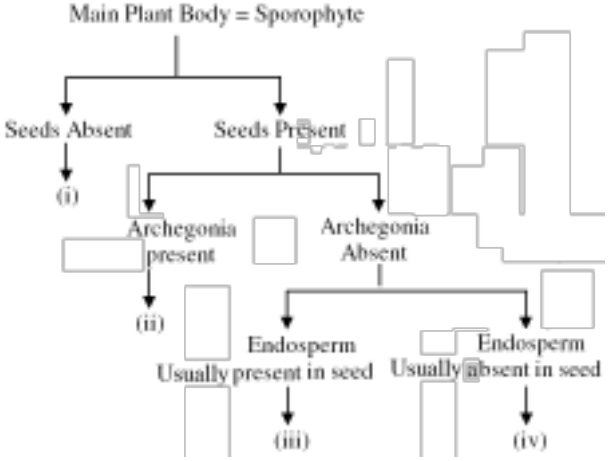
68) Approach of HGP focused on identifying all the genes that expressed as RNA is known as

(1) Sequence annotation
(2) Bioinformatics
(3) Expressed sequence tags
(4) Polymerase chain reaction

69) Ciliated epithelium mainly present in the inner surface of hollow organs like .....A..... and .....B.....
.
Fill in the blanks with A and B :-

(1) A-Bronchioles, B-Fallopian tube
(2) A-Blood Vessel, B-Skin
(3) A-Bone, B-Cartilage
(4) A-Bronchioles, B-Blood Vessel

70) Observe the given table. Fill in the boxes (i), (ii), (iii) and (iv) with appropriate words and select the correct option.



	(i)	(ii)	(iii)	(iv)
(1)	Bryophytes	Angiosperms	Gymnosperms	Pteridophytes
(2)	Pteridophytes	Gymnosperms	Monocots	Dicots
(3)	Gymnosperms	Pteridophytes	Monocots	Dicots
(4)	Pteridophytes	Gymnosperms	Dicots	Monocots

(1) 1

(2) 2

(3) 3

(4) 4

71) All living members of this class are ectoparasite on some fishes. They have a circular & sucking mouth without jaw. Scales & paired fins are absent. 6-15 pair of gill slits present. Identify the animal of this class who have above mentioned characters.

(1) Petromyzon

(2) Amphioxus

(3) Herdmania

(4) Balanoglossus

72) Match column-A with column-B and select the correct option.

Column-A		Column-B	
a	Bacterial disease	i	Diphtheria
b	Fungal disease	ii	Plague
c	Protozoan disease	iii	Pneumenia
		iv	Malaria
		v	Ringworm
		vi	Amoebiasis

(1) a-i, ii; b-iii, v; c-iv, vi

(2) a-ii, iii; b-i, v; c-iv, vi

(3) a-i, ii, iii; b-v; c-iv, vi

(4) a-i, ii, iii; b-iv, v; c-vi

73) With regard to hormones, identify the correct statements :
(A) Gastrin acts on the gastric glands and inhibit the secretion of hydrochloric acid and pepsin.
(B) Epinephrine is a amino acid deriatives hormone.
(C) Estradiol is a peptide hormone.
(D) GIP inhibits gastric secretion and motility.
Choose the most appropriate answer from the options given below :

(1) C and E only

(2) C and D only

(3) B and D only

(4) A and B only

74) Which one is correct for scientific name of mango ?

(1) *Mangifera Indica* Linn

(2) *Mangifera indica Linn*

(3) *mangifera indica* linn

(4) *Mangifera indica* Linn

75) Given below are two statements:
Statement-I: At the point of fusion of the ilium, ischium and pubis a cavity is present called acetabulum to which the thigh bone articulates.
Statement-II: The two halves of pelvic gridle meet dorsally to form the pubic symphysis containing fibrous cartilage.
Choose the correct answer from the options given below:-

(1) Both statement-I and statement-II are correct

(2) Both statement-I and statement-II are incorrect

(3) Statement-I is correct but statement-II is incorrect

(4) Statement-I is incorrect but statement-II is correct

76) A student observe many thread like structures inside the nucleus of cell. This structure become condensed during M-Phase of cell cycle & made up of genetic material and protein. This structure is :

(1) Microtuble

(2) Spindle fiber

(3) Chromatin fiber

(4) Nucleolus

77) How many in the following are the examples of Ex-situ conservation and insitu conservation respectively?
National Park, Seed Bank, Wild Life Safari, Cryopreservation Zoological Parks, Botanical garden,

Biosphere reserve, Wild life sanctuary, Sacred grove.

(1) 3 & 6
(2) 4 & 5
(3) 2 & 7
(4) 5 & 4

78)

Artificial passive acquired immunity is obtained by

(1) IgG through placenta
(2) IgA through colostrum
(3) Antitoxin
(4) Administration of killed pathogen

79) Match List-I with List-II :

List-I [Type of cell]		**List-II [Size]**	
(A)	Mycoplasma	I.	7.0 µm
(B)	Human RBC	II.	10-20 µm
(C)	Bacteria	III.	0.3 µm
(D)	Typical eukaryotic cell	IV.	3 to 5 µm

Choose the correct answer from the options given below :
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-I, C-II, D-IV
(4) A-I, B-II, C-III, D-IV

80) The water vascular system in Echinoderms helps in all of the following EXCEPT:

(1) Locomotion
(2) Capture and transport of food
(3) Respiration
(4) Excretion of nitrogenous waste via kidneys

81) **Statement-I:** The menstrual cycle starts with menstrual phase, when menstrual flow occurs and it lasts for 3-5 days. The menstrual flow results due to breakdown of endometrial lining of the uterus and its blood vessels which forms liquid that comes out through vagina.
Statement-II: During follicular phase, the primary follicles in the ovary grow to become a fully mature graafian follicle and simultaneously the endometrium of uterus regenerates through proliferation. These change in the ovary and the uterus are induced by change in the level of pituitary and ovarian hormone.

(1) Statement-I is true and Statement-II is false

(2) Statement-I is false and Statement-II is true

(3) Both Statement-I and Statement-II are false

(4) Both Statement-I and Statement-II are true

82) Which one is **incorrect** for differentiation?

(1) During differentiation, cells undergoes few to major structural changes.

(2) Structural changes in cells occur in their cells walls and protoplasm.

(3) Cells, which from tracheary elements develop lignocellulosic secondary cell walls .

(4) Cells increase their protoplasm during differentiation to form a tracheary element.

83) Which of the following muscles helps in increasing the volume of the thoracic cavity during inspiration?

(1) Diaphragm

(2) Abdominal muscles

(3) Internal intercostal muscles

(4) Ciliary muscles

84) Loosely attached vitamins with enzyme would be :-

(1) Co-substrate

(2) Co-enzyme

(3) Prosthetic group

(4) Apo-enzyme

85) Find out the incorrect statement :

(1) Secretions of epididymis, vas deference, seminal vesicle and prostate are essential for maturation and motility of sperms.

(2) The seminal plasma along with sperms constitute the semen.

(3) The function of male sex accessory ducts and glands are maintained by the sertoli cells.

(4) The acrosome is filled with enzymes that help fertilization of the ovum.

86) Select the incorrect statement -

(1) Mammals have the ability to produce a concentrated urine.

(2) Each kidney has nearly one million complex tubular structures called nephrons.

(3) A fall in GFR can activate the JG cells to release renin.

(4) An adult human excretes, on an average 25-30 mg of urea per day.

87) **Statement-I :** Adrenal medullary hormones can also increase the cardiac output.
Statement-II : Neural signals through the sympathetic nerves can decrease the heart rate.
In the light of above statements, choose the most appropriate answer from the options below-

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

88) Which of following modification are necessary for most eukaryotic RNA processing :
(a) Addition of unusual nucleotide at 5' end of hnRNA
(b) Cleavage of long precursor RNA into smaller one
(c) Formation of 3' poly A tail in hnRNA
(d) Ligating the intron
Select the correct statement

(1) a, b, c, d

(2) a, b, c

(3) a, c, d

(4) a, c

89) **Assertion (A) :** In PCR reaction, multiple copies of the gene (or DNA) of interest is synthesised in vitro using two sets of primes (small chemically synthesised oligonucleotides that are complementary to the regions of DNA) and the enzyme DNA polymerase.
Reason (R) : The separated DNA fragments can be visualised only after staining the DNA with a compound known as ethidium bromide following by exposure to U.V. radiation.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

90) Which of the following is **not** a functional aspect of an ecosystem?

(1) Productivity

(2) Decomposition

(3) Stratification

(4) Energy flow

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	2	4	4	4	2	3	4	1	3	1	3	3	1	1	3	3	3	1	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	2	2	3	3	1	1	1	1	2	1	2	1	3	3	4	1	4	1	1
Q.	41	42	43	44	45															
A.	3	3	3	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	4	4	2	4	1	2	2	1	3	4	3	3	1	4	3	3	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	1	1	4	1	2	4	2	1	1	3	4	3	3	1	4	4	2	3	2
Q.	86	87	88	89	90															
A.	3	2	1	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	3	1	1	3	3	4	2	3	1	3	3	3	3	1	4	1	3	4	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	2	2	2	3	3	1	3	1	1	2	2	1	4	4	1	4	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	2	4	1	2	2	2	3	2	2	4	3	3	1	4	1	2	1	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	2	2	2	2	2	2	3	3	1	2	1	3	3	4	3	3	4	3	1	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	4	1	2	3	4	3	2	2	3										

SOLUTIONS

PHYSICS

1)

$\lambda_\alpha = \lambda_{Be}$

As $\lambda \propto \frac{1}{P}$

$\Rightarrow P_\alpha = P_{Be}$

$\Rightarrow \sqrt{2M_\alpha V_\alpha q_\alpha} = \sqrt{2M_{Be} V_{Be} q_{Be}}$

$\Rightarrow 4m_P \times 100\ 2e = 8m_P \times V_{Be} \times e$

$\Rightarrow V_{Be} = 100$ Volt

2)

Let displacement of boat is x (towards left)

Their is no external force applied on the system $\Sigma m\Delta x = 0$

$M_R (2 - x) = M_A (2 + x) + M_b (x)$

$60 (2 - x) = 50 (2 + x) + 40 (x)$

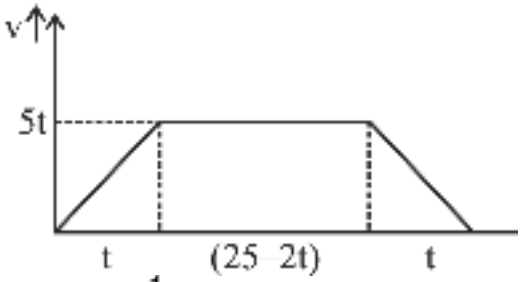
$120 - 60x = 100 + 50x + 40x$

$150x = 120 - 100$

$x = \frac{20}{150} = \frac{2}{15} m$

3) $v = u + at$

$v = 5t$



$v_{av} = \frac{2 \times \frac{1}{2}\left(5t^2\right) + (25 - 2t)5t}{25}$

$20 = \frac{5t^2 + 125t - 10t^2}{25}$

$500 = 125t - 5t^2$

$t^2 - 25t + 100 = 0$

$(t - 20)(t - 5) = 0$

$t = 20, 5$

$= 25 - 2t = 25 - 2(5) = 15$ s

4)

When axis passes through CM, MOI will be minimum.

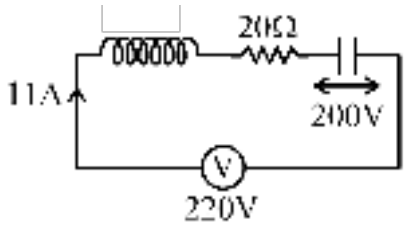
Therefore $\frac{dI}{dx} = 0$

$\Rightarrow 4x - 12 = 0$

$\Rightarrow x = 3$

5)

Due to magnus effect a spinning ball deflects from its original path.



6)

$\because V_R = 11 \times 20 = 220 = V$

$\because V_L = V_C$ = Resonance

= 200 V

7) Impulse, $I = F \times \Delta t = 50 \times 10^{-5} \times 3 = 1.5 \times 10^{-3}$ N – s

8)

Value of reaction

$12 \times 7.68 - 13 \times 7.48 = 92.16 - 97.24 = -5.08$

$\therefore$ Required energy = 5 MeV

9) $\frac{1}{\lambda} = Rz^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\frac{1}{\lambda} = R3^2\left[\frac{1}{1^2} - \frac{1}{3^2}\right] = 8R$

$\lambda = \frac{1}{8R} = \frac{912\,\text{Å}}{8} = 114\,\text{Å}$

$\lambda = 11.4$ nm

10)

Work function is minimum energy required to eject an electron from metal.

11) (i) $E = \frac{\lambda}{2\pi\varepsilon_0 r}$ i.e. $E \propto \frac{1}{r}$ ((i) → (b)) (ii) Inside electric field, E = 0 i.e. $E \propto r^0$ ((ii) → (d))

(iii) $E = \frac{\rho r}{3\varepsilon_0}$ i.e. $E \propto r$ ((iii) → (c))

(iv) $E = \frac{\sigma}{2\varepsilon_0}$ i.e. $E \propto r^0$ ((iv) → (d))

12)

For small angle prism
$\delta = (\mu - 1)A$
correct
(a) $\mu\uparrow \rightarrow \delta\uparrow$
(c) $\delta \propto A$

13)

$V_P - V_Q = I\,R_{eq.} = (0.5)\,(6) = 3V$

14)

Initial slope = $\frac{I_0}{\tau}$ or $\frac{V}{L}$
as from A to B, initial slope decreases, hence L increase with E & R constant.

15) There will be no effect of first zener diode because of forward bias, so
current through 10 kΩ = reading of ammeter

$$= \frac{(20-15)\,V}{10k\Omega} = 0.5mA$$

16)

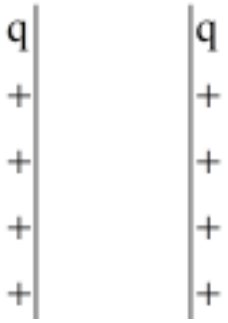
For coherenty frequency of both source must be same and have constant phase difference.

17) $U = \frac{f}{2}\mu RT$

$$U = \frac{5}{2}(3RT) + \frac{3}{2}(5RT) = 15RT$$

18) $C = \frac{\in_0 A}{d}$

As both plates are given same charge, there is no charge on inner surface.



Potential difference V = 0

19)

$y_1 = 10 \sin\left(8\pi t + \frac{\pi}{3}\right)$

$y_2 = 5\left[\sqrt{2}\sin\left(8\pi t + \frac{\pi}{4}\right)\right]$

$\Delta\phi = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12}\text{rad}$

20) $\vec{N} = |\vec{A}|\hat{B} = \frac{5(4\hat{i} + 3\hat{j})}{5} = 4\hat{i} + 3\hat{j}$

$\Rightarrow x = 4$

21) $a = g(\sin\theta - \mu\cos\theta) = 9.8\ (\sin 45^\circ - 0.5\cos 45^\circ)$

$= \frac{4.9}{\sqrt{2}}\text{m/sec}^2$

22) Magnetic moment

$\vec{M} = -i\pi(0.5)^2\hat{k} + i\pi(0.3)^2\hat{k}$

$\vec{M} = -7 \times \frac{22}{7}\left(\frac{25}{100} - \frac{9}{100}\right)\hat{k}$

$= -22\left(\frac{16}{100}\right)\hat{k}$

$\vec{M} = -3.52\hat{k}\ \text{Am}^2$

$= -\frac{7}{2}\hat{k}\ \text{Am}^2$

23) This will happen when, $\frac{1}{f_1} = \frac{1}{f_2}$

$(\mu - 1)\left(\frac{1}{R_1} - \frac{1}{-R_2}\right) = (\mu - 1)\left(\frac{2}{R}\right)$

$\frac{1}{R_1} + \frac{1}{R_2} = \frac{2}{R}$

24) $y = \sqrt{\sin\sqrt{x}}$

$\frac{dy}{dx} = \frac{1}{2\sqrt{\sin\sqrt{x}}} \times \cos\sqrt{x} \times \frac{1}{2\sqrt{x}}$

$= \frac{\cos\sqrt{x}}{4\sqrt{x\sin\sqrt{x}}}$

25) $\vec{E}$ and $\vec{B}$ are mutually perpendicular as well as perpendicular to propagation of electromagnetic wave. But speeds of all electromagnetic waves are same in vacuum.

26) The total work done W is equal to the change in kinetic energy $\Delta K = \frac{1}{2}mv^2 - 0$.
First, find the velocity ν as a function of time t

$\nu = k\sqrt{S} \Rightarrow \frac{dS}{dt} = kS^{1/2} \Rightarrow S^{-1/2}\,dS = kdt$

Integrating from t = 0 to $t : [2S^{1/2}]_0^S = [kt]_0^t \Rightarrow 2\sqrt{S} = kt$

$\sqrt{S} = \frac{kt}{2}$

Velocity $\nu = k\sqrt{S} = k\left(\frac{kt}{2}\right) = \frac{k^2}{2}t$

Work done

$W = \frac{1}{2}mV^2 = \frac{1}{2}m\left(\frac{k^2}{2}t\right)^2 = \frac{mk^4t^2}{8}$

27) In relativity, mass increases with speed as $m = \frac{m_0}{\sqrt{1 - v^2/c^2}}$. But electric charge is a conserved scalar quantity and remains unchanged irrespective of motion. Therefore, unlike mass, it does not depend on velocity.
Hence, the moving particle's charge is always equal to its rest charge : q = q_0.

28) For parallel combination of resistances

$\frac{\Delta R_{eq}}{R_{eq}^2} = \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}\right)$

or $\boxed{\text{Absolute Error } \Delta R_{eq} = \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}\right) R_{eq}^2}$

Here $R_{eq} = \frac{(100)(200)}{(100+200)} = \frac{200}{3}\Omega = 66.7\Omega$

$\Delta R_{eq} = \left(\frac{3}{(10^4)} + \frac{4}{(4\times 10^4)}\right)(66.7)^2$

$= \left(\frac{16}{4\times 10^4}\right)(4445) \cong 1.8\Omega$

Hence R_{eq} = (66.7 ± 1.8)Ω

29) At balance condition $\frac{\ell_1}{\ell_2} = \frac{R}{5} \Rightarrow \frac{AC}{CB} = \frac{3}{4}$

AC + CB = 350 cm

AC = $\frac{3}{7} \times 350$ = 150 cm

30) $I = \frac{I_0}{2}\cos^2\theta$

31)

VT^{-2} = constant

$\Rightarrow PV^{\frac{1}{2}}$ = constant

Q = ΔU + ΔW

$$Q = \frac{nR\Delta T}{\gamma - 1} + \frac{nR\Delta T}{1 - x}$$

$$= \frac{nR50}{\frac{7}{5} - 1} + \frac{nR50}{1 - \frac{1}{2}}$$

= 225R

32) 1. Define the total fixed distance between object and screen as d. Let the lens be at two positions separated by a distance x, with object distances x and v such that:

$$u = \frac{d - x}{2},\ v = \frac{d + x}{2}$$

2. From the lens equation, $f = \frac{d^2 - x^2}{4d}$

3. The magnifications at the two positions are: $m_1 = \frac{v}{u} = \frac{d + x}{d - x}, m_2 = \frac{u}{v} = \frac{d - x}{d + x}$

4. The difference of magnifications is: $m_1 - m_2 = \frac{x}{f}$

5. Rearranging gives: $f = \frac{x}{m_1 - m_2}$

33) Polarization proves the transverse nature of electromagnetic waves.

34) Let 'v' is the volume of metal ball

As half of the ball sinks

 Volume of ice melted is v/2

$Q_{\text{Released by ball}} = Q_{\text{absorbed by ice}}$

$Ms\Delta T = mL_f$

$$4.5 \times v \times (0.1)\,(T - 0) = \frac{v}{2} \times 0.9 \times 80$$

$0.45\ T = 0.45 \times 80$

$T = 80°C$

35) $\alpha = \frac{d\omega}{dt}$

$\alpha = -3t^2 \Rightarrow \frac{d\omega}{dt} = -3t^2$

$$\Rightarrow \int_{27}^{0} d\omega = -\int_{0}^{t} 3t^2 dt \Rightarrow 0 - 27 = -\frac{3t^3}{3}$$

$\Rightarrow t^3 = 27 \Rightarrow t = 3s$

36) $\overline{\overline{A} + \overline{B}} = \overline{A} + \overline{B} = \overline{A \cdot B}$

37) $h \propto \cos\theta$

$$\frac{6}{\cos\theta} = \frac{4}{\cos 60^\circ}$$
$$\cos\theta = \frac{3}{4}$$

38)

$k\Delta = Mg$ .....(1)
Maximum extension = 2Δ
$P.E._{max} = \frac{1}{2}k(2\Delta)^2 = 2k\Delta^2 = 2Mg\Delta$

39) $W_{net} = W_{AB} + W_{BC}$
$W_{AB} = 0$ Isochoric process
$W_{BC} = 2 \times R \times 600 \times \ln\left(\frac{9}{3}\right)$
$W_{BC} = 1200\ R(\ln 3)$

40)

$\omega = 2\pi \times \frac{2}{\pi} = 4$ rad/s
$T \sin\theta = m\omega^2 \ell \sin\theta$

$T = 0.1 \times 16 \times 1 = 1.6$ N

41)

$$B = \frac{\mu_0 I}{4\pi a}(\sin 60^\circ - \sin 30^\circ)$$
$$= \frac{\mu_0}{4\pi}\cdot\frac{I}{a}\cdot\frac{\left(\sqrt{3}-1\right)}{2}$$
$$= \frac{\mu_0 I}{8\pi a}\left(\sqrt{3}-1\right)$$

42)

$a = -2.5t + 10$
$$\frac{dv}{dt} = -2.5t + 10$$
$$\int_{v_0} dv = \int (-2.5t + 10)\,dt$$
$$0 = -\frac{2.5t^2}{2} + 10t$$
$$10t = \frac{2.5}{2}t^2$$
$$\frac{20}{2.5} = t$$

$$\frac{20 \times 10}{25} = t$$

$$\boxed{t = 8 \text{ sec}}$$

43) Let R_0 = Resistance of wire at 0°C

$\Rightarrow 6 = R_0 [1 + \alpha (100–0)]$ ..(1)

$5 = R_0 [1 + \alpha (50–0)]$ ..(2)

from (1), (2) $\Rightarrow \frac{6}{5} = \frac{1 + 100\alpha}{1 + 50\alpha} \Rightarrow \alpha = \frac{1}{200}\,^{\circ}C^{-1}$

put in (1) $6 = R_0 \left[1 + \frac{1}{200} \times 100\right]$

$\Rightarrow R_0 = 4\Omega$

44) $\ell_1 + e = \frac{\lambda}{4} = \frac{V}{4n} = \frac{336}{4 \times 504}$

Here e = 0.3 D

e = 0.3 × 6 = 1.8 cm

$\ell_1 + 1.8 = 16.6$

$\ell_1 = 14.8$ cm

45) $U = \frac{-GMm}{R} = -54$

$\frac{GM}{R} = \frac{54}{m} = \frac{54}{3} = 18$

$V_e = \sqrt{\frac{2GM}{R}} = \sqrt{36} = 6$ m/s

$V_e = 6$ m/s

CHEMISTRY

46) Heat absorbed; q = 150 cal

= 150 × 4.2 = 630 Joule

Work done on the system;

W = +200 Joule

From first law of thermodynamics

ΔU = q + W = 630 + 200 = 830 Joule

47)

Answer Explanation:

Option 1: Lysine, Arginine, and Histidine are basic amino acids → Correct.

Option 2: Deficiency of Vitamin B_2 (Riboflavin) causes cheilosis (cracks at mouth corners) → Correct.

Option 3: Deficiency of Vitamin B_6 (Pyridoxine) causes convulsions → Correct.

Option 4: Deficiency of Vitamin B_2 causes Beri-beri → Incorrect.

Beri-beri is caused by Vitamin B_1 (Thiamine) deficiency, not B_2.D)
Final Answer: Option (4) → Deficiency of Vitamin B_2 does not cause Beri-beri.

48) At equivalence point,
$(N \times V)_{\text{for } KMnO_4} = (N \times V)_{\text{for Mohr's salt}}$
[This question is based experiment based on titration of Mohr's salt v/s $KMnO_4$]

49) NCERT (old) Pg. # 121, Q.No. 4.21

$$k \times t = 2.303 \times \log \frac{P_0}{2P_0 - P_t}$$

50) NCERT (OLD)-XII, Part-1, Pg. # 107, Equation 4.13
From interval formula for 1st order

$$k = \frac{2.303}{(t_2 - t_1)} \times \log_{10} \frac{[A_1]}{[A_2]}$$

$$k = \frac{0.693}{t_{1/2}}$$

After solving we get; $t_{1/2} \simeq 30$ sec.

51) $2VO + 3Fe_2O_3 \rightarrow 6FeO + V_2O_5$

Initial moles $\frac{6.7}{67} = 0.1$ $\frac{4.8}{160} = 0.03$

L.R.T. = $\frac{0.1}{2} = 0.05$ ⇓ L.R.T. = $\frac{0.03}{3} = 0.01$ minimum value

Limiting reagent test
∴ Fe_2O_3 is limiting reagent
Now, from balance reaction

$$\frac{\text{moles of } Fe_2O_3}{3} = \frac{\text{moles of FeO}}{6}$$

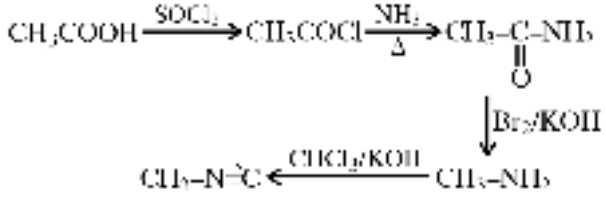
52) $r_{n_2} = 52.9 \times \frac{n_2^2}{1} P_m = 1322.5 \text{ PM}$

$\boxed{n_2 = 5}$

$r_{n_1} = 52.9 \times \frac{n_1^2}{1} P_m = 211.6 \text{ PM}$

$\boxed{n_1 = 2}$

Transition 5 → 2
∴ Balmer series

53) 

54) NCERT-XI, part solved problem 7.24

NH_4OH + HCl ⇌ NH_4Cl + H_2O

At t = 0; moles moles 0 0

= M × V(in litre) = 0.1 × 0.25

= 0.1 × 0.5 = 0.025

= 0.05

After reaction

(0.05 – 0.025) (0.025 – 0.025) 0.025 0.025

= 0.025 = 0 mole mole

In remaining mixture weak base and salt of same weak base with strong acid both are present. Hence it will act as basic buffer solution

$$pOH = pK_b + \log_{10}\frac{[Salt]}{[W.B.]}$$

55)

Cu^+ is unstable in aq. solution and undergoes disproportionation.

$Cu^+(aq.) \rightarrow Cu^{+2}(aq.) + Cu$

56)

Order of IE ⇒ F > Cl > Br > I

57)

Fact based

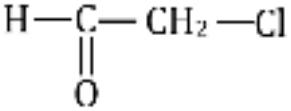
58) **Question Explanation:** Question is asking about the compound which is the most reactive towards S_N2 reaction.

Concept : Bimolecular nucleophilic substitution reaction.

Solution:

Rate of S_N2 reaction ∝ electrophilicity of R – X bond

$$\propto \frac{-M,\ -I}{\text{steric crowding}}$$ around R – X.

Among the given options  is the most reactive towards S_N2 reaction because of –I effect of carbonyl group.

Final Answer: The correct option is (3).

59)

(A) Lower member of aldehyde and ketone are water soluble because they form H-bond with H_2O

This statement is correct.

(B) Boiling point of carbonyl compound lies b/w the boiling point of their correspondent ether and Alcohol.

order of Boiling point ⇒ Alcohol > carbonyl compounds > ether

This statement is correct.

(C) Racemic mixture is formed while new chiral centre is generated during nucleophilic addition reaction of carbonyl compound.

(D) Formalin is aqueous solution of equimolar amount of formic and acetic acid.

It is incorrect because formalin is defined an aqueous solution containing approximately 37% HCHO by mass. A, B & C are correct statement.
Final Answer :- (1)

60)

Phenyl benzoate $\xrightarrow{\text{DIBAL–H}}$ Benzaldehyde.

61) From Arrhenius equation

$k = Ae^{-\frac{E_a}{RT}}$

where $e^{-\frac{E_a}{RT}}$ represent fraction of molecules having energy ≥ E_a

62) $P_2H_4 \xrightarrow{-10e^-} [P_2H_4]^{+10}$

[Compound Y]

In compound Y;

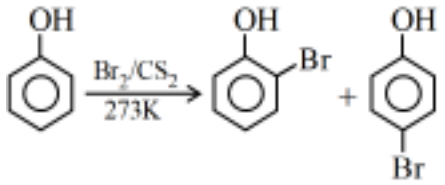
$2x + 4 = +10$

$2x = +6 \Rightarrow [x + 3]$

63) $K_{P_1} = \alpha_1^2 P_1$ and $K_{P_2} = 4\alpha_2^2 P_2$

64)

On the basis of Z_{eff} and number of shells.

65) 

66)

Use molecular orbital diagram.

67) Kjeldahl's method for estimation of nitrogen is not applicable for nitrobenzene C6H5NO2. because reaction with H2SO4, nitrobenzene can not give ammonia.

68) Conductivity = $\frac{1}{R} \times \left(\frac{\ell}{A}\right)$

$\left(\frac{\ell}{A}\right)$ = cell constant = $4 \times 10^{-2} \times 75 = 3\ cm^{-1}$

$\Lambda_m = \frac{1}{R} \times \frac{\ell}{A} \times \frac{1000}{M}$

$\Lambda_m = \frac{1}{250} \times 3 \times \frac{1000}{0.05}$

$= 240\ \Omega^{-1}.cm^2.mol^{-1}$.

69)

T□ shows both +1 and +3 oxidation states in its compounds.

70) Acetone is not oxidised by Tollen's reagent but acetaldehyde oxidised by Tollen's reagent.

71)

I ⇒ $[Co(NH_3)_6]Cl_3$,
II ⇒ $[Co(NH_3)_5Cl]Cl_2$,
III ⇒ $[Co(NH_3)_4Cl_2]Cl$
Order of conductivity ⇒ I > II > III

72)

Al does not react with conc. HNO_3 due to formation of protective layer of oxide on surface.

73) NCERT-XII, Part-I, Pg. # 71

$Mg_{(s)} \rightarrow Mg^{+2} + 2e^-$ (at anode)

$2Ag^+ + 2e^- \rightarrow 2Ag_{(s)}$ (at cathode)

$Mg_{(s)} + 2Ag^+ \rightarrow Mg^{+2} + 2Ag_{(s)}$

$$Q = \frac{[Mg^{+2}]}{[Ag^{+}]^2} = \frac{0.1}{(0.001)^2} = \frac{10^{-1}}{(10^{-3})^2} = \frac{10^{-1}}{10^{-6}} = 10^5$$

$$E_{cell} = E^0_{cell} - \frac{0.06}{2}\log_{10} 10^5$$

$$E_{cell} = 3.17 - \frac{0.06}{2} \times 5\log_{10} 10$$

$= 3.17 - 0.03 \times 5$

$= 3.17 - 0.15 = 3.02$ V

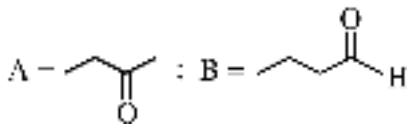
74)

$[Ni(H_2O)_6]^{2+} \Rightarrow Ni^{+2}$ has $t_{2g^6}eg^2$ configuration
$\mu = 2.8$ B.M.
$[Ni(en)_3]^{2+} \Rightarrow Ni^{+2}$ has $t_{2g^6}eg^2$ configuration
$\Rightarrow \mu = 2.8$ B.M.

75) For isotonic solution
$i_1C_1 = i_2C_2$

76)

Cis$[Co(NH_3)_2(en)_2]^{3+}$ does not have plane of symmetry.

77) 

78) This question is based on Hess's law

$Cr_2O_{3(s)} \rightarrow 2Cr_{(s)} + \frac{3}{2}O_{2(g)}$; $\Delta H_1 = +1130$ kJ

$3C_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow 3CO_{(g)}$; $\Delta H_2 = -110 \times 3 = -330$ kJ

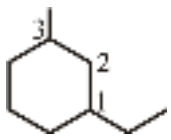
on adding these reactions

$Cr_2O_{3(s)} + 3C_{(s)} \rightarrow 2Cr_{(s)} + 3CO_{(g)}$; $\Delta H_{net} = +1130 - 330$

$= +800$ kJ

79)

ClO_3^- and SO_3^{-2} both have pyramidal shape

80) 

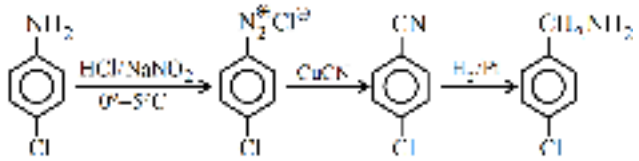
$\Rightarrow$ 1-ethyl-3-methylcyclohexane.

81)

NCERT-XI, Pg.340

82)

ClF_3 and XeF_4 are planer and in XeF_2 all 2 fluorine atoms are in same plane.

83) 

84)

Fact based

85) $$\text{Molality} = \frac{1000\,M}{1000 \times d - M \times (\text{mol.wt.})_{\text{solute}}}$$

or $$3 = \frac{1000M}{1110 - M \times 40}$$

or 3330 – 120 M = 1000 M

$$M = \frac{3330}{1120} = 2.97 \text{ mol/lit}$$

86)

NCERT, Pg.# 347

87)

$NH_3 < PH_3 < AsH_3 < SbH_3 < BiH_3$; Reducing nature

88)

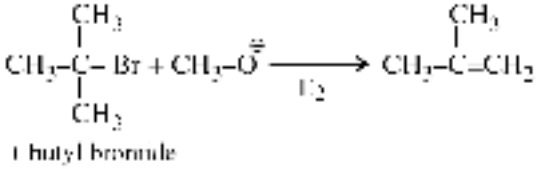
Order of μ ⇒ $CH_3Cl > CH_2Cl_2 > CHCl_3$

$CH_3Cl > CH_3F$

89)

Solution :-

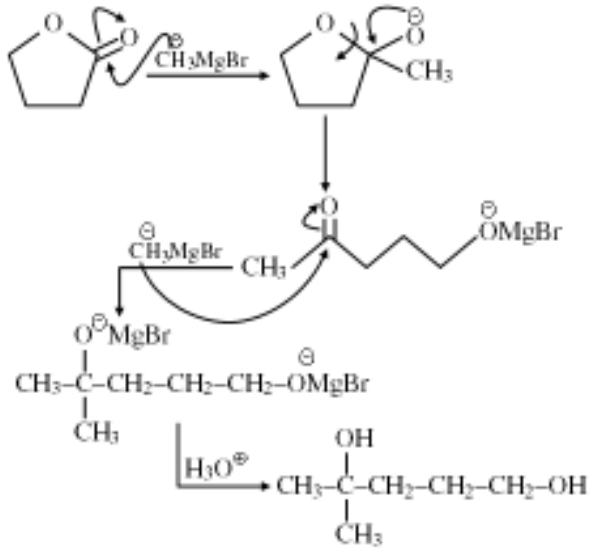
It is crowded substrate. So S_N2 is not possible here, So, E_2 occurs.



t-butyl bromide

Final Answer :- (4)

90)



Final Answer :- Option (1)

BIOLOGY

91) NCERT, Pg. # 1151, 152, 153(E)/166, 167,169(H)

92) NCERT-XII, Pg.# 219-221

93) NCERT Pg. # 127 & 128

94) NCERT, Pg.# 219

95)

NCERT-XII, Pg. # 94

96) NCERT Pg # 126

97)

NCERT Pg. # 180, 181, 182

98)

NCERT-XII, Pg. # 88, 91, 99

99) NCERT Pg.# 35

100) NCERT pg no # 48

101) NCERT-XII, Pg. # 93

102) NCERT XI Pg. # 24,29, 32

103) NCERT pg no # 33

104) NCERT_XII_Pg. No. 235

105) NCERT-XII, Pg. # 61

106) NCERT-XI, Pg. # 149, 150

107) NCERT, Pg. # 40

108) NCERT, Pg # 243

109) NCERT, Pg. # 124-125

110) NCERT Pg.# 179

111) NCERT-XII, Pg # 14

112) NCERT-XII, Pg. # 9, 19, 22, 23

113) NCERT-XI Pg#17,18

114) NCERT-XII, Pg. # 191

115) NCERT Pg. # 33

116) NCERT Pg. # 233

117) NCERT-XII, Pg. # 209

118) NCERT-XI, Page No. 160

119) NCERT Pg.#227

120)

NCERT-XI, Pg. # 145-147

121) NCERT-XI, Pg. # 108

122)

NCERT-XII, Pg. # 59

123) NCERT-XI, Pg. # 162

124) NCERT-XII, Pg. # 212, 213, 214

125) NCERT XI, Pg # 76

126) NCERT-XI, Pg. No.#59.

127)

NCERT-XII, Pg. # 73

128) NCERT, Pg. # 49,50

129) NCERT Pg. # 16

130) NCERT-XII, Pg. # 202

131) NCERT-XI, Page # 175, 176

132) NCERT Pg. # 213

133) NCERT Pg # 99

134)

NCERT-XI, Pg. # 111, 112

135) NCERT-XI, Pg. # 76, 77

136)

NCERT-XII, Pg. # 67

137) NCERT Pg. No.# 65.

138) NCERT Pg.# 17-18

139) NCERT Pg.# 216

140)

NCERT-XII, Pg. # 89

141) NCERT Page No.# 29, 32

142) NCERT, Pg. # 135

143) NCERT Page No. 83

144) NCERT Pg. # 197

145) NCERT Pg. # 169

146) NCERT-XI, Pg. # 151

147) NCERT XI Pg. # 158, 159

148) NCERT XI Pg # 126

149) NCERT, Pg. # 244

150) NCERT-XII, Pg. # 73, 74, 75

151) NCERT, Pg # 151,152,153

152) NCERT, Pg. # 186-187

153) NCERT-XII, Pg. No. # 22, 23

154) Module

155) NCERT (XII) ; Page No. # 199

156) NCERT (XI) Page No. 64

157) NCERT Pg. # 166, 168, 170

158)

NCERT-XII, Pg. # 103

159) NCERT XII Pg. # 101

160) NCERT-XII, Pg. # 31, 32, 33, 34

161) NCERT, Pg. # 47

162) NCERT, Pg. # 131-133

163) NCERT Pg # 247, 248

164) NCERT XI, Pg. # 5

165) NCERT-XIth, Pg. # 226

166) NCERT Page No. # 101.

167) NCERT XII, Pg. # 224, 225

168) NCERT Pg. # 136

169) NCERT Pg # 89, 90

170) NCERT Pg. # 45

171) NCERT pg no # 34

172) NCERT-XI, Pg. # 172

173) NCERT Pg. # 186-187

174)

NCERT-XI, Pg. # 118

175) NCERT Pg. # 32

176) NCERT Pg.#207,211,213

177) NCERT Pg. # 202

178)

NCERT-XII, Pg. # 95

179)

NCERT Pg. # 168 & 172

180) NCERT-XII, Pg. # 206